Every Layer Counts: An Exponential L_2 Depth Hierarchy for ReLU Networks

Itay Safran
Stein Faculty of Computer and Information Science, Ben-Gurion University of the Negev
safrani@bgu.ac.il

Abstract

We prove a depth hierarchy for ReLU neural networks in which every additional ReLU layer can save exponentially many neurons. For every $\ell \geq 3$, a globally $[0, 1]$-valued, 1-Lipschitz function is realized by a depth-ℓ network of width $\mathcal{O}(d^4)$, whereas every depth-$(\ell - 1)$ network with unrestricted weights and width at most $\frac{2^d}{2d(\ell-2)}$ has squared L_2 error at least $1/24$ under an absolutely continuous distribution. To the best of our knowledge, this is the first exponential separation for ReLU networks between two fixed depths whose shallower depth is at least 3, and the first exponential hierarchy across all adjacent fixed depths. The lower bound also immediately yields the corresponding hierarchy for exact computation. Moreover, the case $\ell = 3$ gives a compactly supported depth-3-versus-depth-2 separation with unrestricted shallow-network weights, answering a question raised by Safran et al. [23, Sec. 2.3]. The corresponding distribution nevertheless has all its mass at exponential radius, so the construction falls outside the regularity regime in which such a separation would imply major threshold-circuit lower bounds.

We also prove an exact separation for a more benign function. It is computed by a polynomial-width depth-4 network, whereas every depth-3 network agreeing with it on the unit hypercube requires exponentially many neurons in its first hidden layer, again without any restriction on the weights. The function is globally $[0, 1]$-valued and $\mathcal{O}(\sqrt{d})$-Lipschitz, and maps the unit hypercube onto $[0, 1]$.

1 Introduction

Depth is a defining feature of modern neural-network architectures. Beyond its empirical importance, depth can fundamentally increase representational efficiency: a function admitting a compact deep representation may require a polynomially or even exponentially larger network at a smaller depth. Two seminal works established complementary forms of this phenomenon. Eldan and Shamir [6] proved an exponential separation of depth 3 from depth 2, whereas Telgarsky [26] constructed highly compositional functions separating networks whose depth grows with the problem parameters from substantially shallower networks. Subsequent work broadened these two regimes and progressively clarified the roles played by the dimension, approximation accuracy, target regularity, approximation domain, and weight magnitude [5, 22, 14, 30, 23, 25]. The resulting literature nevertheless remained divided between two regimes: exponential separations of depth 3 from depth 2, and exponential separations between a fixed shallow depth and a deeper architecture whose depth grows with the dimension or the desired accuracy. Recent work established a superlinear depth hierarchy for the exact computation of the maximum function [20], but did not yield an exponential separation. Prior to this work, no exponential separation was known between any two fixed depths $k' > k \geq 3$ for ReLU networks or, to the best of our knowledge, for networks using any other standard nonpolynomial activation.

The analogous question has long been central in circuit complexity. Threshold circuits are essentially feedforward neural networks with threshold activations, operating on Boolean inputs and producing Boolean outputs. A central goal in this setting is a full depth hierarchy: for every constant k, one would like to exhibit an explicit function computable by a polynomial-size depth-$(k+1)$ threshold circuit but requiring exponential size at depth k. Even substantially weaker goals remain open. A separation of depth 3 from depth 2 is known when the weights of the depth-2 circuit are polynomially bounded [10], but the corresponding result for arbitrary weights is unknown. Separating any larger constant depth from depth 3 is likewise a long-standing open problem. Starting at depth 4, candidate pseudorandom functions computable by threshold circuits also place sufficiently general lower-bound techniques within the scope of the natural-proofs barrier [18, 27].

The contrast between the two models is particularly striking because neural networks already admit separations whose threshold-circuit analogues remain open. The construction of Telgarsky [26] separates networks of polynomially growing depth from constant-depth networks, whereas the corresponding separation of polynomial-depth threshold circuits from constant-depth threshold circuits is unknown. Similarly, the depth-3-versus-depth-2 lower bound of Eldan and Shamir [6] places no restriction on the weights of the depth-2 network, while the corresponding arbitrary-weight separation for threshold circuits remains open. Nevertheless, the central fixed-depth gap was shared by both models. For a decade following the results of Eldan and Shamir [6] and Telgarsky [26], no exponential separation was known once the shallower depth was at least 3, even when the two depths were allowed to differ by more than one layer.

Vardi and Shamir [27] formalized the connection between this gap and the corresponding circuit-complexity barriers. Under certain boundedness and regularity assumptions on the target function and input distribution, they showed that, for every $k \geq 4$, an exponential separation of some larger constant ReLU depth from depth k would imply a separation of some larger constant threshold-circuit depth from depth $k-2$. Thus, the case $k=4$ would resolve the arbitrary-weight depth-2 threshold-circuit problem; the case $k=5$ would separate a larger constant depth from depth 3; and the cases $k \geq 6$ would enter the regime governed by the natural-proofs barrier [18]. These implications serve both as motivation and as a boundary marker: a sufficiently regular constant-depth hierarchy for ReLU networks would resolve problems extending far beyond neural-network approximation.

Our main result resolves the fixed-depth gap for ReLU networks and, more strongly, establishes a complete exponential hierarchy across all adjacent depths.

Theorem 1.1 (Informal main theorem)**.** *For every fixed depth $\ell \geq 3$, there is a family of functions represented by polynomial-width depth-ℓ ReLU networks such that approximating them with error below a fixed constant at depth $\ell - 1$ requires width exponential in the input dimension, even with unrestricted weights.*

To compare the normalizations of different L_2 separation results, suppose that the approximation distribution is supported in a ball of radius R, the target is L-Lipschitz there, and the lower bound on the root-mean-square error is ε, namely,

$$\inf_N \sqrt{\mathop{\mathbb{E}}_{\boldsymbol{x}}\left[(N(\boldsymbol{x}) - f(\boldsymbol{x}))^2\right]} \geq \varepsilon.$$

Multiplying the target by a scalar trades the Lipschitz constant against ε, while dilating the input trades the Lipschitz constant against the support radius [23, Theorem 9]. Both operations leave the quantity

$$\frac{RL}{\varepsilon}$$

unchanged. We call its growth the *normalization scale* of the separation. This coarse measure records whether the hardness is accompanied by a large domain, rapid variation of the target, or a demand for unusually fine approximation, without depending on the units chosen for the input or output.

For the formal construction used to prove our main result, we have $L = 1$, $\varepsilon = 1/\sqrt{24}$, and

$$R = 2^{\Theta((\ell-2)d^3)},$$

so the normalization scale is exponential. This is precisely the feature that places our hierarchy outside the barrier of Vardi and Shamir [27]; the large support radius is part of the complexity profile of the result rather than an incidental artifact. Classical depth-3-versus-depth-2 constructions exhibit polynomial dependence in the analogous Lipschitz, radius, or accuracy scales, whereas Safran et al. [25] keeps all three constant. The arbitrary-weight lower bound of Eldan and Shamir [6] instead uses a heavy-tailed distribution and hence has no finite support radius. That construction nevertheless places non-negligible approximation mass at a polynomial distance from the origin. Indeed, its hard target is supported in a ball of radius $\mathcal{O}(\sqrt{d})$, whereas all the mass in our construction lies at exponential radius; see Remark 1.2. By contrast, Telgarsky [26] works on the unit interval but uses a target whose Lipschitz constant grows exponentially with the composition depth. Thus, different separation results place their normalization scale in different parameters.

It remains open whether the normalization scale of our L_2 hierarchy can be reduced to a polynomial, or even a constant, while retaining the unrestricted-weight lower bound. Since the Lipschitz and accuracy scales in our theorem are already constant and the other relevant regularity properties are retained, reducing its support radius to a polynomial would bring the result into the regime of the Vardi–Shamir reduction. Any such improvement would therefore either have to overcome the corresponding circuit-complexity barrier or relax a different assumption.

Remark 1.2 (Compact support and unrestricted weights)**.** *The case $\ell = 3$ answers the question, raised by Safran et al. [23, Sec. 2.3] and restated by Safran et al. [25], of whether an exponential depth-3-versus-depth-2 separation can hold with respect to a compactly supported distribution without any restriction on the shallow network's weights. The latter work obtained a separation on the unit ball with constant Lipschitz and accuracy scales, but under an exponential upper bound on the shallow network's weights. Thus, it achieved optimal normalization while leaving open the simultaneous combination of compact support and unrestricted weights. Our hierarchy imposes no restriction on the weights at any depth, and its $\ell = 3$ specialization gives such an exponential lower bound under an absolutely continuous, compactly supported distribution.*

This conclusion cannot be obtained by simply truncating the heavy-tailed distribution of Eldan and Shamir [6]: its tail probability decreases only polynomially with the truncation radius, while an unrestricted ReLU network may have arbitrarily large linear growth. Its squared error may therefore be concentrated entirely in the discarded tail. Establishing hardness directly under compact support is consequently a genuine strengthening, rather than a truncation of the earlier result. At the same time, the distinction between compactness and scale is important. The hard target of Eldan and Shamir [6] is supported at polynomial radius, whereas the compact support in our construction lies at exponential radius. Thus, our result resolves the compact-support and unrestricted-weight requirements simultaneously, but leaves open the quantitatively stronger goal of obtaining the same separation with radius polynomial in d, or ideally bounded by an absolute constant.

Exact computation provides an alternative setting for studying the expressiveness of neural networks. In this setting, the approximation scale is absent. The maximum function is a central benchmark for this question, as it is easily computed by a logarithmic-depth tournament [1]. Its exact representation, and that of more general continuous piecewise-linear (CPWL) functions, has motivated a growing line of work [16, 11, 9, 3, 2, 8, 4]. Recent work established a superlinear depth hierarchy for the exact computation of the maximum [20], but no exponential fixed-depth lower bound. Our second result gives an exponential separation between depths 4 and 3 for a different, globally bounded CPWL target with controlled geometry.

Theorem 1.3 (Informal exact-computation theorem)**.** *There is a family of functions, one in each dimension, that can be computed exactly by polynomial-width depth-4 ReLU networks, but for which every depth-3 network agreeing with the target on the unit hypercube requires first-layer width exponential in the dimension, even with unrestricted weights.*

For comparison across exact-computation results, let Ω be contained in a ball of radius R, let the target be L-Lipschitz on Ω, and suppose that its image on Ω is an interval of length $I > 0$. Input dilation trades R against L, while output scaling changes L and I by the same factor. Hence the corresponding scale-invariant normalization is

$$\frac{RL}{I}.$$

In our exact-computation result, we may take $R = \sqrt{d}$, $L = 2\sqrt{d}$, and $I = 1$, so the exact-computation normalization scale is at most $2d$. Thus, unlike the approximation hierarchy, this result simultaneously keeps the domain radius and Lipschitz constant polynomially bounded and the image interval non-degenerate. To the best of our knowledge, it is the first exponential lower bound on the width required by depth-3 ReLU networks for exact computation in this regime. It remains open whether every CPWL function in d dimensions can be represented by a depth-3 ReLU network. Our result shows that, even if the answer is affirmative, this simple family has polynomial-width depth-4 representations while every depth-3 representation requires exponential first-layer width.

The remainder of the paper is organized as follows. After reviewing related work below, we introduce notation and conventions in Section 2. Section 3 proves the adjacent-depth L_2 hierarchy, and Section 4 proves the exact depth-4-versus-depth-3 separation. Section 5 summarizes our results and discusses directions for future work. The formal proofs are deferred to the appendices.

1.1 Related work

Separating depth 3 from depth 2 The continuous L_2 literature contains several depth-3-from-depth-2 separations, distinguished by the geometry of the target, the approximation distribution and normalization, and whether the shallow network's weights are restricted. Eldan and Shamir [6] proved the first exponential separation using an approximately radial, oscillatory target and a Fourier analysis argument. Their lower bound permits unrestricted weights, but uses a heavy-tailed distribution and, when expressed with constant Lipschitz and radius scales, an inverse-polynomial accuracy. Venturi et al. [29] extended this Fourier-analytic approach beyond radial functions to product targets and distributions, with analogous heavy-tail and normalization costs. Daniely [5] instead used spherical harmonics to obtain a compactly supported separation on a product of spheres, under an exponential upper bound on the shallow network's weights and again with an inverse-polynomial accuracy after normalization. Reductions and refinements of these analytic methods yielded lower bounds for ball and ellipsoid indicators, other non-oscillatory radial targets, and targets that can be learned by deeper networks [22, 23, 21, 17]. Most recently, Safran et al. [25] obtained an exponential separation on the unit ball with constant Lipschitz and accuracy scales, also under an exponential bound on the shallow network's weights. Complementarily, Hsu et al. [12] showed that, under the uniform distribution on $[-1, 1]^d$, every $\mathcal{O}(1)$-Lipschitz target admits a polynomial-width depth-2 approximation at any fixed constant accuracy, underscoring the role of the approximation distribution. The

$\ell = 3$ case of Theorem 3.1 instead permits unrestricted weights and retains compact support, at the cost of placing that support at exponential radius and concentrating the mass on a union of widely separated cubes.

Despite their different settings, the preceding exponential lower bounds are all tailored to ruling out a single hidden nonlinear layer, and none has yielded an exponential lower bound once the shallower network has depth greater than 2. Our proof instead introduces a new recursive mechanism that reduces hardness at one depth to a hard instance at the preceding depth. Because the same reduction can be applied once per layer, it yields separations at every pair of adjacent depths rather than only between depths 3 and 2.

Deeper separations A second line of work obtains exponential benefits from depth by allowing the depth of the efficient network to vary with the problem parameters. The iterated-sawtooth construction of Telgarsky [26] is computable by constant-width networks with $\Theta(k^3)$ layers, but cannot be approximated at substantially smaller depth $\mathcal{O}(k)$ without $\Omega(2^k)$ neurons. Related region-counting and approximation-theoretic arguments show that depth can efficiently generate the linear pieces needed to approximate smooth or piecewise-smooth functions. Concurrent work by Liang and Srikant [14], Yarotsky [30], and Safran and Shamir [22] established related depth advantages for smooth, piecewise-smooth, and other natural targets, contrasting fixed-depth networks with networks whose depth grows with the desired accuracy and obtaining improved dependence on the inverse accuracy. Further growing-depth separations and approximation-rate results were obtained by Arora et al. [1] and Yarotsky [31], respectively. These results provide strong separations between fixed-depth and growing-depth regimes, but not exponential separations between two fixed depths.

Most recently, Krishnan and Mossel [13] obtained an all-depth tradeoff for exact computation on Boolean inputs. Their product-map family is computed by a constant-width ReLU network of depth $n + 1$, whereas every depth-r, width-w network computing it exactly satisfies $w^r = \Omega(2^n)$. Consequently, polynomial width is ruled out when $r = o(n/\log n)$. This result is again fundamentally a growing-depth separation and, moreover, concerns exact, infinite-accuracy computation on a Boolean domain rather than average-case approximation under a continuous distribution. At the adjacent depth $r = n$, the displayed tradeoff gives only a constant lower bound on the width, and it does not separate any two fixed depths. Thus, neither this result nor the earlier constructions discussed here supply an exponential separation between fixed depths $k' > k \geq 3$. Theorem 3.1 fills precisely this gap, and does so simultaneously for every adjacent pair.

Exact computation Exact representation of CPWL functions removes the approximation parameter, but unconditional depth lower bounds remain elusive when the width and the real-valued weights are unrestricted. Mukherjee and Basu [16] showed that the maximum of three coordinates cannot be represented at depth 2, while Hertrich et al. [11] initiated a systematic study of whether the classes of CPWL functions representable with unrestricted width strictly grow with depth. Subsequent lower bounds for the maximum and for general CPWL functions have required additional assumptions, such as integral or rational weights, monotonicity or input-convexity, or compatibility with a prescribed polyhedral fan [9, 2, 3, 8]. On the upper-bound side, the conjecture of Hertrich et al. [11] that the maximum of five coordinates is not representable at depth 3 was refuted by Bakaev et al. [4], who constructed such a representation and improved the general depth bound for the maximum of d coordinates.

Very recent work pushes this upper-bound frontier considerably further. Rueß et al. [19] proved that the maximum of d coordinates is representable with two hidden layers for every $d \leq 10$ and, through the generalized hinging-hyperplane representation, that every CPWL function on $\mathbb{R}^d$ has such a representation for $d \leq 9$. Their structured representation of the maximum of ten coordinates also yields a recursive construction with fewer layers in arbitrary dimension. This progress makes an unrestricted incomputability result at depth 3 still more delicate and directly motivates a quantitative alternative: even if depth-3 networks can represent every CPWL function, how wide must they be? Theorem 4.1 answers this question exponentially for an explicit, benign target; and Theorem 3.1 implies an exponential separation for all adjacent depths under its exponential-radius normalization. These complement the superlinear exact hierarchy for the maximum in Safran [20], which applies through depths of order $\log \log d$ but does not give an exponential separation. Grillo and Montúfar [7] established a generic depth hierarchy: for an open set of parameters, the realized functions admit no shallower representation with generic parameters, regardless of width. This does not exclude nongeneric shallower representations and therefore yields neither an unconditional depth separation nor a quantitative size lower bound. Our proof builds on a first-layer-collapse principle first used by Safran et al. [24] for approximation of the maximum and subsequently refined by Safran [20] for exact computation. Here we deploy the principle differently: local depth-2 hardness forces first-layer activation hyperplanes through exponentially many perturbed hard points, whose affine general position then yields the width lower bound.

Relations to circuit complexity Several works transfer ideas between neural-network lower bounds and Boolean or communication complexity. An early circuit-simulation viewpoint appears in Martens et al. [15], who used simulations by better-understood circuit models to prove a weight-bounded exponential representational lower bound for restricted Boltzmann machines. For ReLU networks on Boolean inputs, Mukherjee and Basu [16] used random restrictions to obtain an almost-linear lower bound for unrestricted-weight networks consisting of one ReLU layer followed by a threshold output, and sign rank to obtain exponential lower bounds for deeper networks with a threshold output under restrictions on their bottom-layer weights. Vardi et al. [28] subsequently used Boolean and communication complexity to prove nearly linear size lower bounds for benign real-valued functions. In the other direction, Safran et al. [25] combined a depth-2 threshold-circuit lower bound with a worst-to-average-case reduction to obtain a continuous L_2 depth separation on the unit ball, under an exponential bound on the shallow network's weights.

The results of Vardi and Shamir [27] explain why our separations do not immediately resolve open problems in circuit complexity. Under their boundedness and regularity assumptions, a strong separation between fixed ReLU depths can be converted into a new lower bound for threshold circuits. Although Krishnan and Mossel [13] use Boolean inputs, their lower bound requires exact, infinite-precision real outputs; exponentially precise truncations are computable by polynomial-size TC^0 circuits and therefore do not yield a threshold-circuit lower bound. Our results fall outside the circuit reductions for two different reasons: the approximation hierarchy uses a distribution supported at exponential radius, violating an assumption of the Vardi–Shamir reduction, while the exact separation requires exact equality with a continuous real-valued function. Thus, neither result implies the currently unknown lower bounds for threshold circuits.

2 Preliminaries and Notation

Conventions. For a positive integer n, let $[n] := \{1, \ldots, n\}$. We denote vectors using bold-faced letters (e.g. $\boldsymbol{x}$). We write $\langle \boldsymbol{x}, \boldsymbol{z} \rangle$ for the Euclidean inner product, $\|\boldsymbol{x}\|_p$ for the ℓ_p norm, and $\|\boldsymbol{x}\|$ for the Euclidean norm. Functions and maps are written in ordinary italic type, while matrices and linear operators are denoted by capital letters. The notation $\mathrm{Lip}(f)$ denotes the Euclidean-to-Euclidean Lipschitz constant of f. For $\boldsymbol{x} = (x_1, \ldots, x_m) \in \mathbb{R}^m$, we write

$$\mathrm{MAX}_m(\boldsymbol{x}) := \max_{i \in [m]} x_i.$$

We use continuous piecewise-linear (CPWL) for a continuous map that is affine on each cell of some finite polyhedral partition of its domain. A genuine activation hyperplane is the zero set of a non-constant affine preactivation. A finite subset of $\mathbb{R}^d$ is in affine general position if every subfamily of size at most $d+1$ is affinely independent. Constants hidden by $\mathcal{O}(\cdot)$ are universal unless stated otherwise.

Neural Networks. For $t \in \mathbb{R}$, the rectified linear unit (ReLU) is

$$[t]_+ \coloneqq \max\{0, t\},$$

and is applied coordinatewise to vector arguments. We consider fully connected, feedforward networks. A depth-ℓ ReLU network $N : \mathbb{R}^{n_0} \to \mathbb{R}^{n_\ell}$ has the form

$$N = A_\ell \circ [\cdot]_+ \circ A_{\ell-1} \circ \cdots \circ [\cdot]_+ \circ A_1,$$

where each $A_j(\boldsymbol{z}) = W_j \boldsymbol{z} + \boldsymbol{b}_j$ is affine. Thus depth counts affine maps, and a depth-2 network has one hidden ReLU layer. The hidden layer widths are $n_1, \ldots, n_{\ell-1}$; the width of the network is their maximum, and its size is their sum. Unless stated otherwise, all weights and biases are arbitrary real numbers.

Approximation and exact computation. For a probability distribution $\mathcal{D}$ on $\mathbb{R}^d$ and scalar-valued functions $f, N : \mathbb{R}^d \to \mathbb{R}$, their squared $L_2(\mathcal{D})$ error is

$$\|N - f\|^2_{L_2(\mathcal{D})} \coloneqq \mathop{\mathbb{E}}_{\boldsymbol{x} \sim \mathcal{D}} \left[(N(\boldsymbol{x}) - f(\boldsymbol{x}))^2 \right].$$

Throughout the paper, L_2 error refers to this squared quantity. For a set $\Omega \subseteq \mathbb{R}^d$, a network N computes f exactly on Ω if $N(\boldsymbol{x}) = f(\boldsymbol{x})$ for every $\boldsymbol{x} \in \Omega$, and it computes f globally if this holds on all of $\mathbb{R}^d$. In the exact-computation setting, we use *computes*, *represents*, and *realizes* interchangeably. For a measurable set A of positive finite volume, $\mathcal{U}(A)$ denotes the uniform probability distribution on A.

3 Adjacent-Depth L_2 Separation

In this section, we introduce the main result of the paper. It establishes a depth hierarchy for all adjacent depths: at every fixed depth, adding one ReLU layer can reduce the width required for constant-error approximation from exponential to polynomial in the dimension.

Theorem 3.1. *There is a universal constant $C > 0$ such that, for all $d \geq 2$ and $\ell \geq 3$, there exist a globally defined CPWL function*

$$f_{\ell,d} : \mathbb{R}^d \to [0, 1]$$

and an absolutely continuous probability distribution $\mathcal{D}^d_\ell$ such that the following hold.

1. *The function $f_{\ell,d}$ is computed globally by a depth-ℓ ReLU network of width at most Cd^4.*
2. *If a depth-$(\ell - 1)$ ReLU network N has width at most*

$$\frac{2^d}{2d(\ell-2)},$$

then

$$\mathop{\mathbb{E}}_{\boldsymbol{x} \sim \mathcal{D}^d_\ell} \left[(N(\boldsymbol{x}) - f_{\ell,d}(\boldsymbol{x}))^2 \right] \geq \frac{1}{24}.$$

For universal constants $c, C > 0$, the construction given in Appendix C has the following additional properties. The target is globally 1-Lipschitz. The distribution is uniform on the union of $2^{d(\ell-2)}$ pairwise disjoint axis-aligned open cubes, each of side length $2 \cdot 4^{\ell-2}$. Its support lies outside a Euclidean ball of radius $2^{c(\ell-2)d^3}$ and is contained in a Euclidean ball of radius $2^{C(\ell-2)d^3}$.

Previously, no exponential lower bound separated two fixed depths $k' > k \geq 3$ for ReLU networks or, to the best of our knowledge, for networks using any other standard nonpolynomial activation. Theorem 3.1 gives such a separation at every pair of adjacent depths, and therefore a complete adjacent-depth hierarchy for ReLU networks.

Relation to the Vardi–Shamir barrier. For the adjacent depths covered by the reduction of Vardi and Shamir [27], the assumption violated by our construction is almost-bounded support. Their reduction requires that, for every inverse-polynomial δ, all but δ of the probability mass lie in a box $[-R, R]^d$ for some $R = \mathrm{poly}(d)$. By contrast, for every fixed $\ell \geq 3$, the entire support of $\mathcal{D}_\ell^d$ lies outside a Euclidean ball of radius

$$2^{c(\ell-2)d^3}.$$

Consequently, for sufficiently large d, no box of polynomial radius contains any mass, and their reduction to threshold-circuit lower bounds does not apply. This is the precise reason our hierarchy avoids the Vardi–Shamir barrier. For completeness, the other relevant conditions are satisfied: the target is $[0, 1]$-valued, and, for almost every fixed value of all but one coordinate, the conditional density of the remaining coordinate is at most $\frac{1}{2}$.

3.1 Construction and proof technique

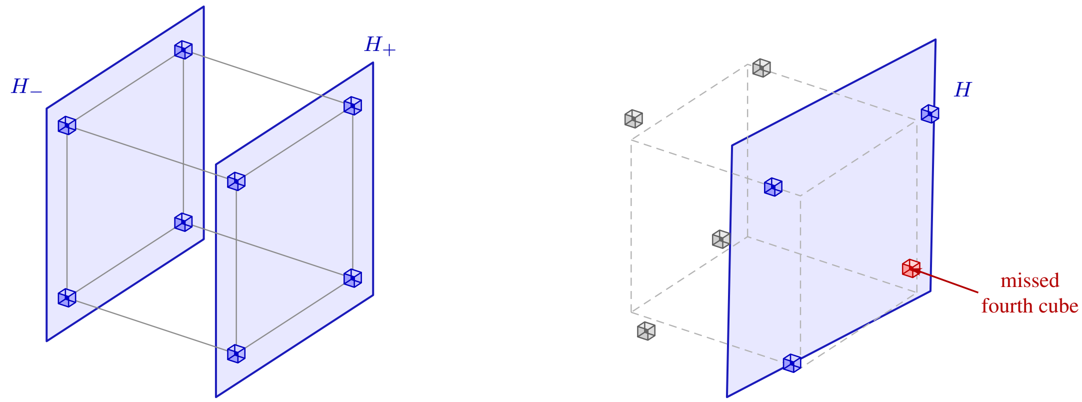
At a high level, our proof idea is based on recursively creating 2^d exact translated-and-rescaled copies of the preceding hard function. A polynomial-width perturbation places their centers in quantitative affine general position, and the copies are made sufficiently small that every affine hyperplane intersects at most d of them. Consequently, the first hidden layer of a narrow network intersects only a small fraction of the copies. On every remaining copy, that layer has a fixed activation pattern and can be merged with the next layer, exposing the preceding hard instance. Iterating this argument yields the hierarchy. After the recursion, a final input rescaling expands all the copies to normalize the target's Lipschitz constant; this expansion is what moves the resulting approximation domain to exponential radius.

The proof has three main steps.

Step 1: Perturbing the Boolean hypercube. Lemma A.1, which will also be used for the exact separation, constructs a uniformly small CPWL displacement Δ and perturbed vertices

$$\boldsymbol{q}_{\boldsymbol{s}} = \boldsymbol{s} + \Delta(\boldsymbol{s}), \qquad \boldsymbol{s} \in \left\{-\frac{1}{2}, \frac{1}{2}\right\}^d.$$

The map Δ is computed by a single-hidden-layer ReLU network of width $\mathcal{O}(d^4)$, while the affine determinant of every $d + 1$ perturbed vertices has magnitude at least $2^{-\mathcal{O}(d^3)}$. This quantitative form of affine general position allows us to place sufficiently small cubes around the vertices so that no affine hyperplane intersects more than d of them. The geometric role of this perturbation is illustrated in Figure 1.



(a) Each of H_- and H_+ meets $4 = 2^{3-1}$ cubes.

(b) The closest plane meeting the three blue cubes still misses the fourth.

Figure 1: The role of the quantitative perturbation in dimension 3. The Boolean cube is rescaled to $\{-1, 1\}^3$, the vertex neighborhoods are enlarged, and the perturbation is exaggerated so that it remains visible at this scale. Before perturbation, each coordinate hyperplane $H_\pm$ intersects $2^{3-1} = 4$ cubes. In the displayed perturbed configuration, no affine plane can intersect more than $d = 3$ cubes. The blue plane attains this bound, the missed fourth cube is red, and cubes irrelevant to this plane are gray. Best viewed in color.

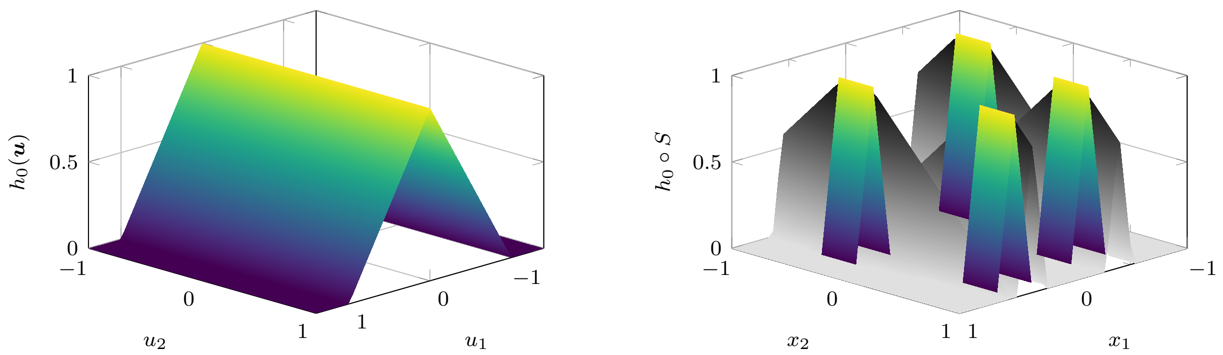
Step 2: Copying a given hard function. Proposition C.2 turns this protected collection of cubes into a copier. For $r = 2^{-\mathcal{O}(d^3)}$, it constructs a depth-2, width-$\mathcal{O}(d^4)$ map S satisfying

$$S(\boldsymbol{q}_{\boldsymbol{s}} + r\boldsymbol{u}) = \boldsymbol{u} \qquad \text{for all } \boldsymbol{s} \in \left\{-\frac{1}{2}, \frac{1}{2}\right\}^d \text{ and } \boldsymbol{u} \in [-1, 1]^d.$$

Consequently, precomposing any function h with S creates 2^d exact translated-and-rescaled copies of h, one inside each protected cube. Indeed,

$$(h \circ S)(\boldsymbol{q}_{\boldsymbol{s}} + r\boldsymbol{u}) = h(\boldsymbol{u}) \qquad \text{for all } \boldsymbol{s} \in \left\{-\frac{1}{2}, \frac{1}{2}\right\}^d \text{ and } \boldsymbol{u} \in [-1, 1]^d.$$

As $\boldsymbol{u}$ ranges over $[-1, 1]^d$, the point $\boldsymbol{q}_{\boldsymbol{s}} + r\boldsymbol{u}$ ranges over the protected cube centered at $\boldsymbol{q}_{\boldsymbol{s}}$, while S recovers the original coordinate $\boldsymbol{u}$ exactly. Thus $h \circ S$ restricts to an exact translated-and-rescaled copy of h on every protected cube. Figure 2 illustrates this identity in dimension 2 for the affine base function $h_0(\boldsymbol{u}) = [1 - |u_1|]_+$ used in the induction.



(a) The affine base function $h_0(\boldsymbol{u}) = [1 - |u_1|]_+$.

(b) Four perturbed copies of h_0, joined by affine pieces.

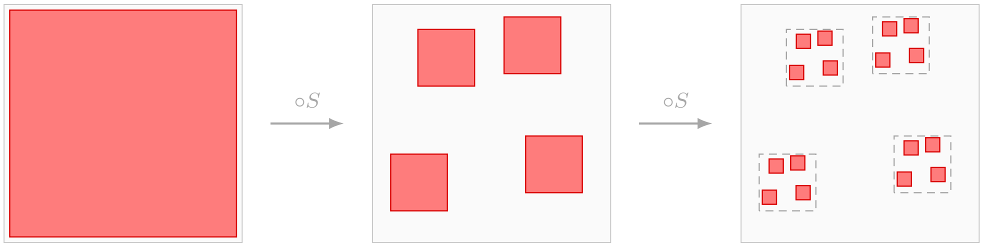
Figure 2: The one-layer copier in dimension 2. The colored patches in Subfigure (b) are the four protected squares $\boldsymbol{q_s} + r[-1, 1]^2$. On each square, the copier identity makes the surface an exact translated-and-rescaled copy of Subfigure (a); the grayscale surface shows the piecewise-affine interpolation outside them. The four centers have been perturbed from the grid $\{-\frac{1}{2}, \frac{1}{2}\}^2$. The copy size and the perturbation are exaggerated for visibility. Best viewed in color.

When h is computed by a ReLU network, the affine output of S merges into the first affine map computing h. Thus the composition $h \circ S$ adds exactly one hidden layer and only $\mathcal{O}(d^4)$ width.

Step 3: Iterating the separation. The induction begins with the base function $h_0(\boldsymbol{x}) = 1 - |x_1|$ on $(-1, 1)^d$, shown in Figure 2a. Lemma C.1 shows that its squared error under the uniform distribution is at least $1/12$ for every affine approximant. After j applications of the copier, the target and its distribution consist of equally weighted copies of the preceding hard instance. Figure 3 gives a bird's-eye view of this recursion in dimension 2. If a depth-$(j + 1)$ network has width w, its first-layer activation hyperplanes intersect at most wd of the 2^d outer copies. On every missed copy, the first layer has a fixed activation pattern and can be merged into the next layer. Applying the induction hypothesis on those copies yields the lower bound

$$\frac{1}{12}\left(1 - \frac{wd}{2^d}\right)^j.$$

Taking $j = \ell - 2$ and the width from Theorem 3.1 makes this quantity at least $1/24$. A final input dilation makes the target globally 1-Lipschitz without changing its network complexity or squared approximation error; this dilation is also what moves the support to exponential radius.



(a) $\mathrm{supp}(\mathcal{D}_0^2)$: 1 square (b) $\mathrm{supp}(\mathcal{D}_1^2)$: 4 squares (c) $\mathrm{supp}(\mathcal{D}_2^2)$: 16 squares

Figure 3: Iterating the copier in dimension 2. Red marks the support, on which the distribution is uniform. Each application replaces every square by four shrunken, perturbed copies, so one square becomes four and then sixteen. The dashed boxes in Subfigure (c) show the parents from Subfigure (b). Sizes and perturbations are exaggerated for visibility. Best viewed in color.

4 An Exact Depth-4-versus-Depth-3 Separation

The hierarchy theorem already implies exact separations, but its 1-Lipschitz normalization places the corresponding hard distribution at exponential radius. We now show that, at adjacent depths 4 and 3, an exact separation is possible while keeping the computation domain, Lipschitz constant, and image scale polynomially controlled.

Theorem 4.1. *There is a universal constant $C > 0$ such that, for all $d \geq 3$, there exists a globally defined CPWL function*

$$g_d : \mathbb{R}^d \to [0, 1]$$

such that

$$\mathrm{Lip}(g_d) \leq 2\sqrt{d}, \qquad g_d([-1,1]^d) = [0,1],$$

and the following hold.

1. *The function g_d is computed globally by a depth-4 ReLU network whose hidden-layer widths are at most Cd^4, $2d$, and 1.*
2. *Every depth-3 ReLU network agreeing with g_d throughout $(-1,1)^d$ has first hidden-layer width at least*

$$\frac{2^{d-1}}{d}.$$

Recall from the introduction that, when the computation domain lies in a ball of radius R, the target is L-Lipschitz there, and its image is an interval of length $I > 0$, we use the scale-invariant normalization RL/I. For the construction below, the computation domain is the unit hypercube $[-1,1]^d$. The cube is contained in the Euclidean ball of radius $R = \sqrt{d}$, the target is at most $2\sqrt{d}$-Lipschitz, and its image is the full interval $[0,1]$. Thus $I = 1$ and

$$\frac{RL}{I} \leq \frac{\sqrt{d}\,(2\sqrt{d})}{1} = 2d,$$

so this normalization is linear in d.

To the best of our knowledge, this is the first exponential lower bound for exact computation in a regime where the domain radius and Lipschitz constant are polynomially bounded while the target image has constant length.[1] The significance of this result is that, even if every CPWL function in d dimensions can be represented by a depth-3 ReLU network, this simple, benign family admits polynomial-width depth-4 representations, whereas every depth-3 representation requires exponential first-layer width.

4.1 Construction and proof technique

The lower-bound mechanism has two ingredients. First, suppose a target has a point at which its restriction to every neighborhood cannot be computed by a depth-2 network. If a depth-3 network computes the target, then a first-layer activation hyperplane must pass through that point. Indeed, if no such hyperplane passes through it, every first-layer ReLU has a fixed activation state on some neighborhood of the point. The first layer is therefore affine on that neighborhood and can be absorbed into the next layer, yielding the forbidden local depth-2 representation. We refer to such a point as a *depth-2 hard point*.

Second, we construct a target with exponentially many depth-2 hard points in affine general position. A hyperplane can contain at most d points in affine general position in $\mathbb{R}^d$. Consequently, covering all the hard points requires exponentially many first-layer activation hyperplanes, and hence exponentially many first-layer neurons.

The construction begins with the simple ℓ_1 function

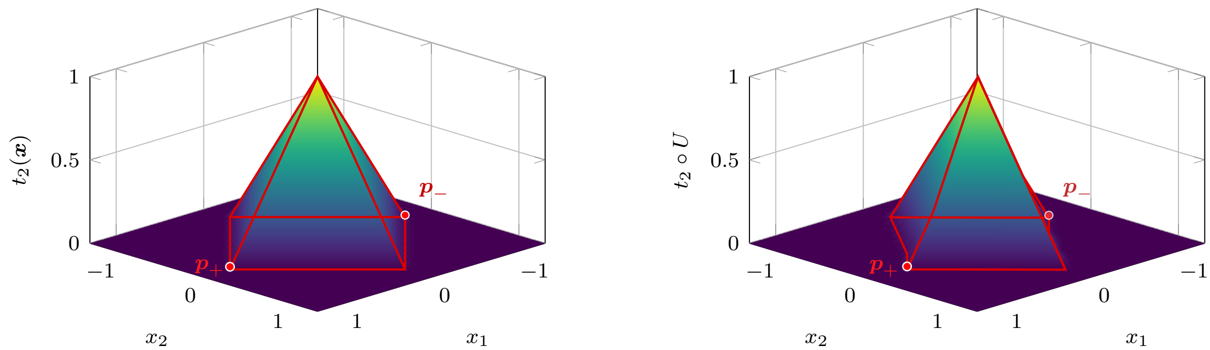
$$t_d(\boldsymbol{x}) \coloneqq [1 - \|\boldsymbol{x}\|_1]_+ ,$$

which has 2^{d-1} depth-2 hard points on its boundary. We then construct a CPWL map $U : \mathbb{R}^d \to \mathbb{R}^d$ that is uniformly close to the identity, moves these points into affine general position, and preserves the local function around every one of them. The separating target is

$$g_d \coloneqq t_d \circ U.$$

Figure 4 illustrates the mechanism in dimension 2. The perturbation moves the hard points, while acting as a translation on a neighborhood of each one. Thus the target's behavior around every transported hard point agrees, after a translation of coordinates, with t_d on a neighborhood of the corresponding original depth-2 hard point.

[1] One can normalize the exponentially growing Lipschitz constant in Telgarsky's iterated sawtooth construction by dilating its input, without affecting exact representability. The required dilation, however, makes the domain radius exponential in the depth parameter, just as the final dilation in Section 3 moves our approximation distribution to exponential radius. Scaling the output instead keeps the domain fixed but shrinks the image interval exponentially. Theorem 4.1 simultaneously keeps the domain radius at $\sqrt{d}$, the Lipschitz constant at $\mathcal{O}(\sqrt{d})$, and the image equal to $[0, 1]$.



(a) The base function t_2. Red curves outline its non-differentiability set, and its two depth-2 hard points are marked in red.

(b) The perturbed target $t_2 \circ U$, with its transported non-differentiability set outlined. The transported $\boldsymbol{p}_+$ and $\boldsymbol{p}_-$ are both marked in red.

Figure 4: A two-dimensional illustration of the exact construction. Subfigure (a) shows the base function with both hard points marked. The red curves outline its non-differentiability set: the boundary $|x_1|+|x_2| = 1$ together with the two interior ridges. Subfigure (b) traces the transported boundary and interior ridges in red. Around each marked point, the perturbation acts as a translation, so the function's local behavior is unchanged. Any additional minor creases created by the illustrative CPWL interpolation for U are omitted from Subfigure (b). The displacement is enlarged for visibility. Best viewed in color.

The figure isolates the role of the perturbation: it moves each hard point while leaving the function unchanged up to translation on a full neighborhood of that point. In dimension d, the 2^{d-1} transported hard points are in affine general position. The first ingredient above forces a first-layer hyperplane through every one of them, while the second allows each hyperplane to cover at most d of them. This gives the lower bound $2^{d-1}/d$. For the upper bound, one hidden layer computes U, after which two further hidden layers compute t_d; together these form the claimed depth-4 network.

5 Conclusions and Future Work

We established a complete exponential hierarchy for ReLU networks across all adjacent depths. For every $\ell \geq 3$, our construction gives a globally bounded and Lipschitz target that is computed by a polynomial-width depth-ℓ network, but whose constant-error approximation at depth $\ell - 1$ requires exponential width, even when the shallower network has unrestricted weights. We also obtained a complementary exact-computation separation between depths 4 and 3 on the unit hypercube, while keeping the target globally bounded and its Lipschitz constant polynomially controlled.

At the level of architectural resources, the approximation hierarchy has the strongest form one could seek: the compared depths are adjacent, and every additional layer yields an exponential reduction in width. The main remaining opportunity is therefore to strengthen the regularity of the separation rather than its depth or width dependence. In our construction, the approximation distribution is compactly supported and otherwise regular, but its support lies at exponential radius. This places the result outside the polynomial-radius, or almost-bounded-support, regime underlying the barrier of Vardi and Shamir [27]. Whether a comparable hierarchy can be established within that regime remains a major open problem.

A particularly natural first step is an exponential L_2 separation of depth 4 from depth 3 satisfying the Vardi–Shamir assumptions. To the best of our knowledge, such a result would not by itself imply any major new threshold-circuit lower bound, unlike analogous separations in which the shallower ReLU network has depth at least 4. The depth-4-from-depth-3 case may therefore be the most plausible setting in which to determine whether the exponential support radius used here is essential or can be replaced by a polynomially bounded, or even constant-radius, approximation domain. Resolving this case would provide the first exponential fixed-depth separation beyond depth 3 versus depth 2 within the full regularity regime, and would clarify how far an adjacent-depth hierarchy can be pushed before the known circuit-complexity barriers become unavoidable.

Acknowledgments

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A Perturbing hypercube vertices into general position

We index the Boolean-cube vertices directly by $\{-\frac{1}{2}, \frac{1}{2}\}^d$; the sign of coordinate s_i is therefore $2s_i$. The following lemma produces a small, polynomial-width displacement field whose perturbed sign vertices have a quantitative affine-determinant margin.

Lemma A.1 (Quantitatively independent perturbations of hypercube vertices)**.** *There is a universal constant $C > 0$ with the following property. Let $d \geq 2$. There are points $\boldsymbol{q_s} \in \mathbb{R}^d$, indexed by $\boldsymbol{s} \in \{-\frac{1}{2}, \frac{1}{2}\}^d$, and a globally defined CPWL map $\Delta : \mathbb{R}^d \to \mathbb{R}^d$ such that:*

1. *Δ is computed by a vector-valued depth-2 ReLU network of width at most Cd^4, and*
$$\boldsymbol{q_s} := \boldsymbol{s} + \Delta(\boldsymbol{s});$$
2. *globally,*
$$\sup_{\boldsymbol{x}\in\mathbb{R}^d} \|\Delta(\boldsymbol{x})\|_\infty \leq \frac{1}{64d}, \qquad \mathrm{Lip}(\Delta) \leq \frac{1}{16};$$
3. *the displacement is locally constant at every perturbed vertex:*
$$\Delta(\boldsymbol{q_s} + \boldsymbol{u}) = \Delta(\boldsymbol{s}) \qquad \textit{whenever } \|\boldsymbol{u}\|_\infty \leq \frac{1}{8d};$$
4. *for all $d+1$ distinct indices $\boldsymbol{s}_0, \ldots, \boldsymbol{s}_d \in \{-\frac{1}{2}, \frac{1}{2}\}^d$, let $Q(\boldsymbol{s}_0, \ldots, \boldsymbol{s}_d) \in \mathbb{R}^{d\times d}$ be the matrix whose i^{th} column is $\boldsymbol{q}_{\boldsymbol{s}_i} - \boldsymbol{q}_{\boldsymbol{s}_0}$. Then*
$$|\det Q(\boldsymbol{s}_0, \ldots, \boldsymbol{s}_d)| \geq 2^{-Cd^3}.$$

In particular, the points $\left\{\boldsymbol{q_s} : \boldsymbol{s} \in \{-\frac{1}{2}, \frac{1}{2}\}^d\right\}$ are in affine general position.

Proof. We first construct a polynomial-size collection of sign vectors such that every $d+1$ of them are linearly independent. For $A \subseteq [d]$ and $\boldsymbol{s} \in \{-\frac{1}{2}, \frac{1}{2}\}^d$, define
$$\chi_A(\boldsymbol{s}) := \prod_{i\in A}(2s_i).$$

Independently sample each of $A_1, \ldots, A_M$ uniformly from the power set of $[d]$. For $\boldsymbol{s} \in \{-\frac{1}{2}, \frac{1}{2}\}^d$, define
$$\boldsymbol{z_s} := (\chi_{A_1}(\boldsymbol{s}), \ldots, \chi_{A_M}(\boldsymbol{s})) \in \{-1, 1\}^M.$$

Fix $\boldsymbol{v} \in \{-1,1\}^d \setminus \{\mathbf{1}\}$ before drawing the subsets, and choose a coordinate i_0 with $v_{i_0} = -1$. For each j, set $Y_j(\boldsymbol{v}) \coloneqq \prod_{i \in A_j} v_i$. Conditional on all membership choices for A_j except whether $i_0 \in A_j$, including i_0 flips the sign of $Y_j(\boldsymbol{v})$. A uniformly random subset includes i_0 with probability $\frac{1}{2}$, independently of all other membership choices. Thus $Y_j(\boldsymbol{v})$ is an unbiased sign, and the variables $Y_1(\boldsymbol{v}), \ldots, Y_M(\boldsymbol{v})$ are independent because the subsets are drawn independently. For each fixed $\boldsymbol{v}$, Hoeffding's inequality bounds each of the upper- and lower-tail events by $\exp\left(-\frac{M}{8d^2}\right)$. There are $2^d - 1$ admissible vectors $\boldsymbol{v} \in \{-1,1\}^d \setminus \{\mathbf{1}\}$. A union bound over both tails and all such vectors therefore gives

$$\mathbb{P}_{A_1,\ldots,A_M}\left[\max_{\boldsymbol{v} \in \{-1,1\}^d \setminus \{\mathbf{1}\}} \left| \frac{1}{M} \sum_{j=1}^{M} Y_j(\boldsymbol{v}) \right| > \frac{1}{2d}\right] \leq 2^{d+1} \exp\left(-\frac{M}{8d^2}\right).$$

Take $M \coloneqq \lceil C_1 d^3 \rceil$ for a sufficiently large universal constant C_1. The displayed probability bound is then strictly below one. By the probabilistic method, there exists a realization of $A_1, \ldots, A_M$ such that

$$\max_{\boldsymbol{v} \in \{-1,1\}^d \setminus \{\mathbf{1}\}} \left| \frac{1}{M} \sum_{j=1}^{M} Y_j(\boldsymbol{v}) \right| \leq \frac{1}{2d}. \tag{1}$$

For any two distinct $\boldsymbol{s}, \boldsymbol{s}' \in \left\{-\frac{1}{2}, \frac{1}{2}\right\}^d$, define $\boldsymbol{v} \in \{-1,1\}^d$ by $v_i \coloneqq 4 s_i s_i'$. Then $\boldsymbol{v} \neq \mathbf{1}$ since $\boldsymbol{s}, \boldsymbol{s}'$ are distinct, and

$$\chi_{A_j}(\boldsymbol{s})\chi_{A_j}(\boldsymbol{s}') = \prod_{i \in A_j} (2s_i) \prod_{i \in A_j} (2s_i') = \prod_{i \in A_j} 4 s_i s_i' = Y_j(\boldsymbol{v})$$

for all j. Therefore,

$$\frac{1}{M} \langle \boldsymbol{z}_{\boldsymbol{s}}, \boldsymbol{z}_{\boldsymbol{s}'} \rangle = \frac{1}{M} \sum_{j=1}^{M} Y_j(\boldsymbol{v}).$$

Thus, by Equation (1), distinct normalized vectors $M^{-\frac{1}{2}} \boldsymbol{z}_{\boldsymbol{s}}$ have inner product of magnitude at most $\frac{1}{2d}$. The Gram matrix of any $k \leq d+1$ of them is strictly diagonally dominant, since

$$\frac{k-1}{2d} \leq \frac{d}{2d} = \frac{1}{2}.$$

Hence every $d+1$ distinct vectors among the $\boldsymbol{z}_{\boldsymbol{s}}$ are linearly independent.

We next realize these signs by a CPWL map that is constant near every sign vertex. Choose a continuous piecewise-linear function $g : \mathbb{R} \to [-1,1]$ which equals $(-1)^j$ on

$$[j - \tfrac{1}{4}, j + \tfrac{1}{4}], \qquad j = 0, \ldots, d,$$

interpolates linearly between consecutive plateaus, and is constant on its two unbounded tails.

The function g has exactly $2d$ breakpoints and can therefore be represented by an affine combination of exactly $2d$ scalar ReLUs. For $A \subseteq [d]$, consider the function

$$\boldsymbol{x} \mapsto g\left(\sum_{i \in A} \left(\frac{1}{2} - x_i\right)\right).$$

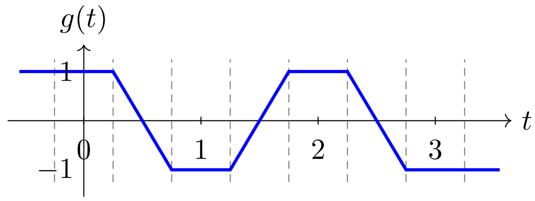

Figure 5: The plateau function g, shown for $d = 3$. Around each integer $j \in \{0, \dots, d\}$, it is constant and equal to $(-1)^j$. Hence a small perturbation of the input to g preserves the parity encoded at that integer; this is what makes the feature map below locally constant around every Boolean-cube vertex.

At $\boldsymbol{x} = \boldsymbol{s} \in \left\{-\frac{1}{2}, \frac{1}{2}\right\}^d$, the sum in the argument is $\sum_{i \in A} \left(\frac{1}{2} - s_i\right)$, the number of negative coordinates of $\boldsymbol{s}$ in A. Consequently,

$$g\left(\sum_{i \in A} \left(\frac{1}{2} - s_i\right)\right) = (-1)^{\sum_{i \in A}\left(\frac{1}{2} - s_i\right)} = \prod_{i \in A} (2s_i) = \chi_A(\boldsymbol{s}).$$

Define the vector-valued map $\widetilde{z} : \mathbb{R}^d \to \mathbb{R}^M$ directly from these plateau functions by

$$\widetilde{z}(\boldsymbol{x}) := \left(g\left(\sum_{i \in A_j} \left(\frac{1}{2} - x_i\right)\right)\right)_{j=1}^{M}.$$

If $\|\boldsymbol{x} - \boldsymbol{s}\|_\infty < \frac{1}{4d}$, then both points fall on the same plateau:

$$\left|\sum_{i \in A} \left(\frac{1}{2} - x_i\right) - \sum_{i \in A} \left(\frac{1}{2} - s_i\right)\right| \le \sum_{i \in A} |x_i - s_i| < \frac{1}{4}.$$

Thus, if $\|\boldsymbol{x} - \boldsymbol{s}\|_\infty < \frac{1}{4d}$, we have

$$\widetilde{z}(\boldsymbol{x}) = \boldsymbol{z}_{\boldsymbol{s}}. \tag{2}$$

Each coordinate of $\widetilde{z}$ is an affine combination of exactly $2d$ scalar ReLUs. Thus all M coordinates are computed by one depth-2 vector-valued ReLU network with at most $2dM = \mathcal{O}(dM) = \mathcal{O}(d^4)$ hidden neurons and an affine output map. Every coordinate is bounded by one. Moreover, each coordinate is $4\sqrt{d}$-Lipschitz, because g has slopes of magnitude at most 4 and the affine function inside it has gradient norm at most $\sqrt{d}$. Summing the squared coordinatewise bounds gives

$$\mathrm{Lip}(\widetilde{z}) \le 4\sqrt{Md}. \tag{3}$$

At this point, $\widetilde{z}$ maps the d-dimensional input into $\mathbb{R}^M$ and is locally constant around every $\boldsymbol{s} \in \left\{-\frac{1}{2}, \frac{1}{2}\right\}^d$, with value $\boldsymbol{z}_{\boldsymbol{s}}$. It remains to map these feature vectors back to $\mathbb{R}^d$ by a fixed linear map whose norm is small and for which the perturbed points become quantitatively affinely independent.

Fix one ordering of each $(d+1)$-subset, write it as $I = (\boldsymbol{s}_0, \dots, \boldsymbol{s}_d)$. Define $X_I \in \mathbb{R}^{d \times d}$ to be the matrix whose i^{th} column is $\boldsymbol{s}_i - \boldsymbol{s}_0$, and define $Z_I \in \mathbb{R}^{M \times d}$ to be the matrix whose i^{th} column is $\boldsymbol{z}_{\boldsymbol{s}_i} - \boldsymbol{z}_{\boldsymbol{s}_0}$. The linear independence just proved implies that Z_I has rank d. Choose a left inverse $L_I \in \mathbb{R}^{d \times M}$, so that

$$L_I Z_I = I_d. \tag{4}$$

A well-chosen matrix $B \in \mathbb{R}^{d\times M}$ maps the plateau features back to $\mathbb{R}^d$ and produces the candidate perturbations $B\boldsymbol{z}_{\boldsymbol{s}}$. We will choose one small rational matrix B below. For a fixed tuple I, the difference matrix of the resulting points $\boldsymbol{s} + B\boldsymbol{z}_{\boldsymbol{s}}$ is $X_I + BZ_I$. We regard the dM entries of B as scalar variables. For $i, j \in [d]$, the corresponding entry of this difference matrix is

$$(X_I + BZ_I)_{ij} = (X_I)_{ij} + \sum_{k=1}^{M} B_{ik}(Z_I)_{kj},$$

which is an affine polynomial in those variables. Accordingly, define

$$D_I(B) \coloneqq \det(X_I + BZ_I).$$

By the determinant expansion, D_I is a polynomial in the entries of B. Its total degree is at most d, because every term in the determinant expansion is a product of d entries of $X_I + BZ_I$. This polynomial is not identically zero. Indeed, substituting $B = (I_d - X_I)L_I$ and using Equation (4) gives

$$X_I + BZ_I = X_I + (I_d - X_I)L_I Z_I = I_d,$$

and hence $D_I((I_d - X_I)L_I) = 1$.

Our goal is to choose one rational matrix B for which $D_I(B) \neq 0$ simultaneously for all $(d+1)$-tuples I. This makes every $d+1$ of the perturbed points $\boldsymbol{s}+B\boldsymbol{z}_{\boldsymbol{s}}$ affinely independent. Two additional quantitative requirements are important. The entries of B must be small enough that the resulting displacement field satisfies the uniform and Lipschitz bounds asserted in the lemma, while their common denominator must be controlled so that a non-zero determinant cannot be arbitrarily small. The latter will yield the inverse-exponential determinant margin in the lemma.

We enforce all the non-vanishing requirements simultaneously by considering the polynomial

$$\Pi(B) \coloneqq \prod_{I} D_I(B). \tag{5}$$

We then find a rational grid point where Π is non-zero; the range of the grid controls the size of B, while its common denominator gives the quantitative determinant lower bound.

There are $\binom{2^d}{d+1}$ possible choices of I. Each determinant D_I is a non-zero polynomial of total degree at most d. Therefore, Π is a non-zero polynomial of total degree at most

$$d\binom{2^d}{d+1} \leq d(2^d)^{d+1} < 2^d 2^{d(d+1)} = 2^{d(d+2)}.$$

Here the first inequality bounds the number of $(d+1)$-subsets by the number of ordered $(d+1)$-tuples, and the strict inequality uses $d < 2^d$. The polynomial Π is non-zero because a finite product of non-zero real polynomials is itself non-zero.

Let

$$L \coloneqq \lceil 64dM \rceil. \tag{6}$$

A non-zero polynomial of total degree at most D cannot vanish on a Cartesian grid S^n when $|S| > D$. We prove this by induction on n. For $n = 1$, this is the elementary univariate root bound: a non-zero polynomial of degree at most D has at most D distinct roots. For $n > 1$, write the polynomial uniquely as

$$F(x_1, \dots, x_n) = \sum_{k=0}^{r} c_k(x_1, \dots, x_{n-1}) x_n^k.$$

Because F is not the zero polynomial, its coefficient polynomials $c_0, \dots, c_r$ cannot all be zero. Choose an index k for which c_k is non-zero. The polynomial c_k need not depend on every one of the first $n-1$ variables; it is simply a polynomial in $x_1, \dots, x_{n-1}$, and its total degree is at most D. By the induction hypothesis, choose $(x_1, \dots, x_{n-1}) \in S^{n-1}$ at which c_k is non-zero. After this substitution, the monomial x_n^k has a non-zero coefficient, so $x_n \mapsto F(x_1, \dots, x_{n-1}, x_n)$ is a non-zero univariate polynomial in x_n of degree at most D. It vanishes on at most D elements of S, so some choice of $x_n \in S$ makes F non-zero.

Apply this fact to the polynomial Π defined in Equation (5), with a grid having $2^{d(d+2)} + 1$ values in each of its dM coordinates. It follows that some matrix

$$B \in \left\{0, \frac{1}{L2^{d(d+2)}}, \dots, \frac{1}{L}\right\}^{d \times M}$$

satisfies $\Pi(B) \neq 0$. If some factor $D_I(B)$ were zero, then $\Pi(B)$ would be zero; hence every factor $D_I(B)$ is non-zero. Thus this single matrix B makes the difference matrix of every $(d+1)$-tuple non-singular, which is precisely the simultaneous affine-independence property sought above. Define

$$\Delta(\boldsymbol{x}) := B\widetilde{z}(\boldsymbol{x}), \qquad \boldsymbol{q}_{\boldsymbol{s}} := \boldsymbol{s} + B\boldsymbol{z}_{\boldsymbol{s}} = \boldsymbol{s} + \Delta(\boldsymbol{s}).$$

For all tuples I, the matrix whose columns are $\boldsymbol{q}_{\boldsymbol{s}_i} - \boldsymbol{q}_{\boldsymbol{s}_0}$ is $X_I + BZ_I$ and therefore has determinant $D_I(B) \neq 0$. Every entry of B lies in $\left[0, \frac{1}{L}\right]$, while every coordinate of $\widetilde{z}(\boldsymbol{x})$ has magnitude at most one. Thus, for all $i \in [d]$ and all $\boldsymbol{x} \in \mathbb{R}^d$,

$$|\Delta_i(\boldsymbol{x})| = \left| \sum_{j=1}^{M} B_{ij} \widetilde{z}_j(\boldsymbol{x}) \right| \leq \sum_{j=1}^{M} |B_{ij}| \, |\widetilde{z}_j(\boldsymbol{x})| \leq \sum_{j=1}^{M} \frac{1}{L} = \frac{M}{L}.$$

Consequently,

$$\sup_{\boldsymbol{x}} \|\Delta(\boldsymbol{x})\|_\infty \leq \frac{M}{L} \leq \frac{1}{64d},$$

where the final inequality follows directly from Equation (6). Moreover, the operator norm of a matrix is at most its Frobenius norm, so

$$\|B\|_{\mathrm{op}} \leq \|B\|_{\mathrm{F}} = \sqrt{\sum_{i=1}^{d} \sum_{j=1}^{M} B_{ij}^2} \leq \frac{\sqrt{dM}}{L}.$$

Combining this estimate with Equation (3) gives a global Lipschitz bound on $\mathbb{R}^d$. Indeed, for all $\boldsymbol{x}_1, \boldsymbol{x}_2 \in \mathbb{R}^d$,

$$\begin{aligned}
\|\Delta(\boldsymbol{x}_1) - \Delta(\boldsymbol{x}_2)\|_2 &\leq \|B\|_{\mathrm{op}} \|\widetilde{z}(\boldsymbol{x}_1) - \widetilde{z}(\boldsymbol{x}_2)\|_2 \\
&\leq \frac{\sqrt{dM}}{L} 4\sqrt{Md} \, \|\boldsymbol{x}_1 - \boldsymbol{x}_2\|_2 \\
&\leq \frac{4dM}{L} \|\boldsymbol{x}_1 - \boldsymbol{x}_2\|_2 \leq \frac{1}{16} \|\boldsymbol{x}_1 - \boldsymbol{x}_2\|_2 .
\end{aligned}$$

The last inequality again uses Equation (6). Therefore, $\operatorname{Lip}(\Delta) \le \frac{1}{16}$ on all of $\mathbb{R}^d$.

Set $R := 2L2^{d(d+2)}$. Every coordinate of every $\boldsymbol{q_s}$ is an integer divided by R. Since $M = \lceil C_1 d^3 \rceil = \mathcal{O}(d^3)$, Equation (6) gives $L = \mathcal{O}(d^4)$ and hence

$$R = 2L2^{d(d+2)} \le 2^{d(d+2)+\mathcal{O}(\log d)} = 2^{\mathcal{O}(d^2)}.$$

Consequently, for all tuples I, the non-zero determinant $D_I(B)$ is an integer divided by R^d and therefore has magnitude at least

$$R^{-d} \ge 2^{-C_2 d^3}$$

for a sufficiently large universal constant $C_2 > 0$. Enlarging the universal constant in the lemma statement proves its determinant bound, which also gives affine general position.

Finally, if $\|\boldsymbol{u}\|_\infty \le \frac{1}{8d}$, then

$$\begin{aligned}
\|\boldsymbol{q_s} + \boldsymbol{u} - \boldsymbol{s}\|_\infty &= \|(\boldsymbol{q_s} - \boldsymbol{s}) + \boldsymbol{u}\|_\infty \\
&\le \|\boldsymbol{q_s} - \boldsymbol{s}\|_\infty + \|\boldsymbol{u}\|_\infty \\
&= \|\Delta(\boldsymbol{s})\|_\infty + \|\boldsymbol{u}\|_\infty \\
&\le \frac{1}{64d} + \frac{1}{8d} < \frac{1}{4d}.
\end{aligned}$$

The plateau identity in Equation (2) now gives

$$\Delta(\boldsymbol{q_s} + \boldsymbol{u}) = B\boldsymbol{z_s} = \Delta(\boldsymbol{s}),$$

as claimed. $\square$

B Proof of Theorem 4.1

Lemma B.1 (Local depth-2 hardness of t_d)**.** *Let $d \ge 3$, and define*

$$t_d(\boldsymbol{x}) := [1 - \|\boldsymbol{x}\|_1]_+ \qquad \textit{and} \qquad \boldsymbol{p_s} := \left(\frac{2\boldsymbol{s}}{d-1}, 0\right), \qquad \boldsymbol{s} \in \left\{-\frac{1}{2}, \frac{1}{2}\right\}^{d-1}.$$

No finite depth-2 ReLU network agrees with t_d on a neighborhood of any $\boldsymbol{p_s}$.

Proof. Each $\boldsymbol{p_s}$ lies on the boundary of the unit ℓ_1 ball because

$$\|\boldsymbol{p_s}\|_1 = \sum_{i=1}^{d-1} \frac{2|s_i|}{d-1} = 1.$$

Fix $\boldsymbol{s}$. Starting at $\boldsymbol{p_s}$, consider moving in a direction $(\boldsymbol{v}, z) \in \mathbb{R}^{d-1} \times \mathbb{R}$ and evaluating t_d at $\boldsymbol{p_s} + (\boldsymbol{v}, z)$. If $\|\boldsymbol{v}\|_\infty < \frac{1}{d-1}$, then, for every $i \in [d-1]$,

$$\operatorname{sign}\left(\frac{2s_i}{d-1} + v_i\right) = \operatorname{sign}\left(\frac{2s_i}{d-1}\right) = \operatorname{sign}(s_i),$$

because $|2s_i/(d-1)| = 1/(d-1) > |v_i|$. Hence

$$\left|\frac{2s_i}{d-1} + v_i\right| = |2s_i|\left|\frac{2s_i}{d-1} + v_i\right| = \left|\frac{4s_i^2}{d-1} + 2s_i v_i\right| = \frac{1}{d-1} + 2s_i v_i.$$

It follows that

$$\begin{aligned}\|\boldsymbol{p_s} + (\boldsymbol{v}, z)\|_1 &= \sum_{i=1}^{d-1} \left| \frac{2s_i}{d-1} + v_i \right| + |z| \\ &= 1 + 2\langle \boldsymbol{s}, \boldsymbol{v}\rangle + |z|,\end{aligned}$$

and therefore, whenever $\|\boldsymbol{v}\|_\infty < \frac{1}{d-1}$,

$$t_d\big(\boldsymbol{p_s} + (\boldsymbol{v}, z)\big) = [-2\langle \boldsymbol{s}, \boldsymbol{v}\rangle - |z|]_+ . \tag{7}$$

Now restrict t_d to the two-dimensional affine plane through $\boldsymbol{p_s}$ parameterized by $t, z \in \mathbb{R}$ by setting

$$\boldsymbol{v} := -\frac{2t}{d-1}\boldsymbol{s}.$$

Since $\langle \boldsymbol{s}, \boldsymbol{s}\rangle = \frac{d-1}{4}$, the restriction of Equation (7) to this plane is $[t - |z|]_+$. Under the invertible linear change of variables

$$t := \frac{x+y}{2}, \qquad z := \frac{x-y}{2},$$

we have

$$t - |z| = \frac{x + y - |x-y|}{2} = \min\{x, y\}.$$

Thus a depth-2 representation of t_d on a neighborhood of $\boldsymbol{p_s}$ would give a local depth-2 representation for $[\min\{x, y\}]_+$. Adding the two ReLU neurons $[x]_+$ and $[y]_+$ would then give a local depth-2 representation of

$$[x]_+ + [y]_+ - [\min\{x, y\}]_+ = \max\{0, x, y\}.$$

This contradicts the two-dimensional $\max\{0, x, y\}$ obstruction in the proof of Proposition A.2 of Safran [20].[2] This proves the lemma. □

Proof of Theorem 4.1. Around each marked point in Lemma B.1, the function t_d has a local form that no finite depth-2 network can represent. These points do not yet have the geometric independence needed for the counting argument. We perturb them into affine general position while preserving the function exactly on a neighborhood of every point.

Apply Lemma A.1 in dimension d. Denote its displacement map by $\widehat{\Delta}$ and its perturbed sign vertices by

$$\widehat{\boldsymbol{q}}_{\boldsymbol{\sigma}}, \qquad \boldsymbol{\sigma} \in \left\{-\frac{1}{2}, \frac{1}{2}\right\}^d,$$

whose unperturbed locations are $\boldsymbol{\sigma}$.

We use only the face indexed by $\left(\boldsymbol{s}, -\frac{1}{2}\right)$, where $\boldsymbol{s} \in \left\{-\frac{1}{2}, \frac{1}{2}\right\}^{d-1}$. Translation by $\frac{\boldsymbol{e}_d}{2}$ sends this face to the hyperplane $x_d = 0$, and scaling by $\frac{2}{d-1}$ sends its unperturbed vertices to the hard points:

$$\frac{2}{d-1}\left(\left(\boldsymbol{s}, -\frac{1}{2}\right) + \frac{\boldsymbol{e}_d}{2}\right) = \left(\frac{2\boldsymbol{s}}{d-1}, 0\right) = \boldsymbol{p_s}.$$

[2] Proposition A.2 is stated as an impossibility result for computing MAX_3 by a depth-2 ReLU network. Its proof establishes the stronger local fact used here: no finite depth-2 ReLU network agrees with $\max\{0, x, y\}$ on any neighborhood of the origin.

Apply the same affine transformation to the perturbed vertices and conjugate the displacement map by this transformation:

$$\begin{aligned}\boldsymbol{q}_{\boldsymbol{s}} &\coloneqq \frac{2}{d-1}\left(\widehat{\boldsymbol{q}}_{(\boldsymbol{s},-\frac{1}{2})}+\frac{\boldsymbol{e}_d}{2}\right),\\ \Delta(\boldsymbol{x}) &\coloneqq \frac{2}{d-1}\widehat{\Delta}\left(\frac{d-1}{2}\boldsymbol{x}-\frac{\boldsymbol{e}_d}{2}\right).\end{aligned} \tag{8}$$

Item 1 of Lemma A.1 and the same change of coordinates also give

$$\boldsymbol{q}_{\boldsymbol{s}} = \boldsymbol{p}_{\boldsymbol{s}} + \Delta(\boldsymbol{p}_{\boldsymbol{s}}). \tag{9}$$

The points $\widehat{\boldsymbol{q}}_{(\boldsymbol{s},-\frac{1}{2})}$ form a subset of the affinely general-position family supplied by the lemma. A common translation and non-zero scaling preserve affine general position, so the 2^{d-1} points $\boldsymbol{q}_{\boldsymbol{s}}$ are also in affine general position. The definition of Δ in Equation (8) preserves the network depth and width and, together with Item 2 of Lemma A.1, gives

$$\sup_{\boldsymbol{x}\in\mathbb{R}^d}\|\Delta(\boldsymbol{x})\|_\infty \leq \frac{1}{32d(d-1)}, \qquad \mathrm{Lip}(\Delta) \leq \frac{1}{16}. \tag{10}$$

Most importantly, Item 3 of Lemma A.1 preserves this local form exactly. If $\|\boldsymbol{u}\|_\infty \leq \frac{1}{4d(d-1)}$, then

$$\frac{d-1}{2}(\boldsymbol{q}_{\boldsymbol{s}}+\boldsymbol{u}) - \frac{\boldsymbol{e}_d}{2} = \widehat{\boldsymbol{q}}_{(\boldsymbol{s},-\frac{1}{2})} + \frac{d-1}{2}\boldsymbol{u}, \qquad \left\|\frac{d-1}{2}\boldsymbol{u}\right\|_\infty \leq \frac{1}{8d}.$$

Using the definition of Δ in Equation (8), the above displayed equation, and Item 3 of Lemma A.1, we obtain

$$\begin{aligned}\Delta(\boldsymbol{q}_{\boldsymbol{s}}+\boldsymbol{u}) &= \frac{2}{d-1}\widehat{\Delta}\left(\widehat{\boldsymbol{q}}_{(\boldsymbol{s},-\frac{1}{2})}+\frac{d-1}{2}\boldsymbol{u}\right)\\ &= \frac{2}{d-1}\widehat{\Delta}\left(\left(\boldsymbol{s},-\frac{1}{2}\right)\right)\\ &= \Delta(\boldsymbol{p}_{\boldsymbol{s}}).\end{aligned} \tag{11}$$

The first and final equalities in Equation (11) use Equation (8); the middle equality uses the local-constancy item of the shared lemma.

Define

$$U(\boldsymbol{x}) \coloneqq \boldsymbol{x} - \Delta(\boldsymbol{x}), \qquad g_d \coloneqq t_d \circ U. \tag{12}$$

Equations (9) and (11), together with the definition of U in Equation (12), imply

$$U(\boldsymbol{q}_{\boldsymbol{s}}+\boldsymbol{u}) = \boldsymbol{p}_{\boldsymbol{s}}+\boldsymbol{u} \qquad \left(\|\boldsymbol{u}\|_\infty \leq \frac{1}{4d(d-1)}\right). \tag{13}$$

Consequently, Equations (12) and (13) give

$$g_d(\boldsymbol{q}_{\boldsymbol{s}}+\boldsymbol{u}) = t_d(\boldsymbol{p}_{\boldsymbol{s}}+\boldsymbol{u}) \tag{14}$$

for all $\boldsymbol{u}$ in this neighborhood of $\mathbf{0}$. Thus the entire local form of t_d around $\boldsymbol{p}_{\boldsymbol{s}}$ is reproduced around $\boldsymbol{q}_{\boldsymbol{s}}$, so no finite depth-2 ReLU network agrees with g_d on a neighborhood of $\boldsymbol{q}_{\boldsymbol{s}}$: otherwise, translating its input and using Equation (14) would contradict Lemma B.1.

All marked points lie in the open unit hypercube. Indeed, Equations (9) and (10) give

$$\|\boldsymbol{q}_{\boldsymbol{s}}\|_\infty \le \|\boldsymbol{p}_{\boldsymbol{s}}\|_\infty + \|\Delta(\boldsymbol{p}_{\boldsymbol{s}})\|_\infty \le \frac{1}{d-1} + \frac{1}{32d(d-1)} < 1.$$

Suppose now that a depth-3 ReLU network N agrees with g_d throughout $(-1,1)^d$. Fix one marked point $\boldsymbol{q}_{\boldsymbol{s}}$. If no genuine first-layer activation hyperplane of N contained this point, then a sufficiently small neighborhood of it would avoid every such hyperplane. After discarding identically zero preactivations, every first-layer neuron would have a fixed activation state there. The first hidden layer would therefore be affine on that neighborhood and could be absorbed into the next affine map. This would produce a local depth-2 representation of g_d, contradicting Lemma B.1 through Equation (14).

Thus every one of the 2^{d-1} marked points lies on a first-layer activation hyperplane. Affine general position permits one hyperplane to contain at most d marked points. If the first hidden width is w, then

$$dw \ge 2^{d-1}.$$

Consequently,

$$w \ge \frac{2^{d-1}}{d}.$$

This proves the shallow lower bound.

It remains to verify the realization and regularity claims. Item 1 of Lemma A.1 and Equation (8) show that the first hidden layer computes the ReLU features whose appropriate affine combinations give the coordinates of Δ. Adding the $2d$ neurons $[x_i]_+, [-x_i]_+$ allows the next affine map to recover each input coordinate as $x_i = [x_i]_+ - [-x_i]_+$. Hence the second hidden layer computes

$$[(U(\boldsymbol{x}))_i]_+, \qquad [-(U(\boldsymbol{x}))_i]_+, \qquad i = 1, \dots, d,$$

and the third hidden layer consists of the single neuron computing

$$\left[1 - \sum_{i=1}^{d} \left([(U(\boldsymbol{x}))_i]_+ + [-(U(\boldsymbol{x}))_i]_+\right)\right]_+ = [1 - \|U(\boldsymbol{x})\|_1]_+ = t_d(U(\boldsymbol{x})) = g_d(\boldsymbol{x}).$$

Here the first equality uses $|a| = [a]_+ + [-a]_+$, the second uses the definition $t_d(\boldsymbol{x}) = [1 - \|\boldsymbol{x}\|_1]_+$, and the final equality uses Equation (12). After increasing the universal constant, the three hidden widths are at most $Cd^4, 2d, 1$. Since Δ is globally CPWL, so is U; the displayed network computes g_d globally, so g_d is also CPWL. This proves the deep upper bound.

The Lipschitz estimate in Equation (10) and the definition of U in Equation (12) give

$$\mathrm{Lip}(U) \le 1 + \frac{1}{16} = \frac{17}{16}.$$

Since $\mathrm{Lip}(t_d) = \sqrt{d}$, the above displayed equation and Equation (12) give

$$\mathrm{Lip}(g_d) \le \frac{17\sqrt{d}}{16} < 2\sqrt{d}.$$

Finally, Equation (10) shows that Δ is a contraction on the complete metric space $\mathbb{R}^d$, since its Lipschitz constant is strictly below 1. The Banach fixed-point theorem therefore gives a unique point $\boldsymbol{x}_0 \in \mathbb{R}^d$ satisfying $\Delta(\boldsymbol{x}_0) = \boldsymbol{x}_0$. By Equation (12), $U(\boldsymbol{x}_0) = \mathbf{0}$. Equation (10) also gives

$$\|\boldsymbol{x}_0\|_\infty = \|\Delta(\boldsymbol{x}_0)\|_\infty \le \frac{1}{32d(d-1)} < 1.$$

Thus $\boldsymbol{x}_0$ lies in $(-1,1)^d$, and Equation (12) yields

$$g_d(\boldsymbol{x}_0) = t_d(U(\boldsymbol{x}_0)) = t_d(\mathbf{0}) = 1. \tag{15}$$

Set $\boldsymbol{y} \coloneqq \mathbf{1}/\sqrt{d}$, so that $\boldsymbol{y} \in [-1,1]^d$ and $\|\boldsymbol{y}\|_1 = \sqrt{d}$. Equations (12) and (10) give

$$\|U(\boldsymbol{y})\|_1 \geq \sqrt{d} - \|\Delta(\boldsymbol{y})\|_1 \geq \sqrt{d} - \frac{1}{32(d-1)} > 1,$$

and hence

$$g_d(\boldsymbol{y}) = t_d(U(\boldsymbol{y})) = [1 - \|U(\boldsymbol{y})\|_1]_+ = 0. \tag{16}$$

Equation (12) and the definition $t_d(\boldsymbol{x}) = [1 - \|\boldsymbol{x}\|_1]_+$ show that $g_d(\mathbb{R}^d) \subseteq [0,1]$. Moreover, g_d is continuous, so the image of the connected cube $[-1,1]^d$ is an interval. By Equations (15) and (16), this interval contains both endpoints of $[0,1]$. Therefore,

$$g_d([-1,1]^d) = [0,1].$$

□

C Proof of Theorem 3.1

Lemma C.1 (The affine base-function gap)**.** *Let*

$$h_0(\boldsymbol{x}) \coloneqq [x_1 + 1]_+ - 2\,[x_1]_+ + [x_1 - 1]_+\,, \qquad \mathcal{D}_0^d \coloneqq \mathcal{U}((-1,1)^d).$$

Then the constant function $\boldsymbol{x} \mapsto \frac{1}{2}$ is an optimal affine approximation to h_0 in $L_2(\mathcal{D}_0^d)$, and

$$\inf_{A \text{ affine}} \mathop{\mathbb{E}}_{\boldsymbol{x} \sim \mathcal{D}_0^d} \left[\big(h_0(\boldsymbol{x}) - A(\boldsymbol{x})\big)^2 \right] = \frac{1}{12}.$$

Proof. On $(-1,1)^d$,

$$h_0(\boldsymbol{x}) = 1 - |x_1|.$$

Globally, $h_0(\boldsymbol{x}) = [1 - |x_1|]_+$, so h_0 takes values in $[0,1]$ and vanishes whenever $|x_1| \geq 1$. Its graph is shown in Figure 2a; it is a triangular ridge that is constant in every direction perpendicular to x_1.

Let $X \sim \mathcal{D}_0^1$. We first reduce the d-dimensional optimization to its one-dimensional counterpart. For an arbitrary affine function $A : \mathbb{R}^d \to \mathbb{R}$, write $\boldsymbol{x}' \coloneqq (x_2, \ldots, x_d)$. Under $\mathcal{D}_0^d$, the first coordinate and $\boldsymbol{x}'$ are independent, with respective laws $\mathcal{D}_0^1$ and $\mathcal{D}_0^{d-1}$. For fixed $\boldsymbol{x}'$, the restriction $x \mapsto A(x, \boldsymbol{x}')$ is affine. Therefore, for any affine A,

$$\mathop{\mathbb{E}}_{\boldsymbol{x}' \sim \mathcal{D}_0^{d-1}} \left[\mathop{\mathbb{E}}_{X \sim \mathcal{D}_0^1} \left[\big(1 - |X| - A(X, \boldsymbol{x}')\big)^2 \right] \right] \\ \geq \inf_{a,b \in \mathbb{R}} \mathop{\mathbb{E}}_{X \sim \mathcal{D}_0^1} \left[\big(1 - |X| - aX - b\big)^2 \right].$$

Taking the infimum over A gives the lower-bound direction. Conversely, for all $a, b \in \mathbb{R}$, the affine function $\boldsymbol{x} \mapsto a x_1 + b$ has loss equal to the corresponding one-dimensional expectation. Taking the infimum over a, b gives the reverse direction. Hence

$$\inf_{A \text{ affine}} \mathop{\mathbb{E}}_{\boldsymbol{x} \sim \mathcal{D}_0^d} \left[\big(h_0(\boldsymbol{x}) - A(\boldsymbol{x})\big)^2 \right] = \inf_{a,b \in \mathbb{R}} \mathop{\mathbb{E}}_{X \sim \mathcal{D}_0^1} \left[\big(1 - |X| - aX - b\big)^2 \right]. \tag{17}$$

It remains to evaluate this one-dimensional infimum. For arbitrary $a, b \in \mathbb{R}$, expanding the squared error gives

$$\begin{aligned}\mathbb{E}\left[(1-|X|-aX-b)^2\right] &= \mathbb{E}\left[(1-|X|-b)^2\right] - 2a\,\mathbb{E}\left[X(1-|X|-b)\right] + a^2\,\mathbb{E}\left[X^2\right] \\ &= \mathbb{E}\left[(1-|X|-b)^2\right] + a^2\,\mathbb{E}\left[X^2\right].\end{aligned}$$

The second equality holds because $1-|X|-b$ is even in X, whereas X is odd, so the mixed expectation vanishes by symmetry. Thus the optimal slope is $a=0$, while the optimal constant is

$$b = \mathbb{E}\left[1-|X|\right] = \frac{1}{2}\int_{-1}^{1}(1-|x|)\,dx = \int_0^1 (1-x)\,dx = \frac{1}{2}.$$

Its one-dimensional squared error is

$$\begin{aligned}\mathbb{E}\left[\left(1-|X|-\frac{1}{2}\right)^2\right] &= \frac{1}{2}\int_{-1}^{1}\left(\frac{1}{2}-|x|\right)^2 dx \\ &= \int_0^1\left(\frac{1}{2}-x\right)^2 dx \\ &= \left.\frac{(x-\frac{1}{2})^3}{3}\right|_0^1 = \frac{1}{12}.\end{aligned}$$

Together with Equation (17), this proves the claimed error, and the constant function $\frac{1}{2}$ attains it. $\square$

Proposition C.2 (The one-layer copier)**.** *There are universal constants $C, C_0 > 0$ such that, for all $d \geq 2$, with*

$$r := 2^{-C_0 d^3},$$

there are centers $\left\{\boldsymbol{q_s} : \boldsymbol{s} \in \left\{-\frac{1}{2}, \frac{1}{2}\right\}^d\right\}$ and a vector-valued depth-2 ReLU map $S : \mathbb{R}^d \to \mathbb{R}^d$ of width at most Cd^4 such that:

1. $$\|\boldsymbol{q_s} - \boldsymbol{s}\|_\infty \leq \frac{1}{64d};$$

2. *the cubes $\boldsymbol{q_s} + [-r, r]^d$ are pairwise disjoint and contained in $(-1,1)^d$, and every affine hyperplane intersects at most d of them;*

3. *for all $\boldsymbol{s} \in \left\{-\frac{1}{2}, \frac{1}{2}\right\}^d$ and $\boldsymbol{u} \in [-1,1]^d$,*

$$S(\boldsymbol{q_s} + r\boldsymbol{u}) = \boldsymbol{u}; \tag{18}$$

4. $\mathrm{Lip}(S) < \frac{4}{r}$.

Consequently, for all functions $h : \mathbb{R}^d \to \mathbb{R}$, all $\boldsymbol{s} \in \left\{-\frac{1}{2}, \frac{1}{2}\right\}^d$, and all $\boldsymbol{u} \in [-1,1]^d$,

$$(h \circ S)(\boldsymbol{q_s} + r\boldsymbol{u}) = h(\boldsymbol{u}).$$

The final identity is the central conclusion: precomposition by one depth-2 ReLU map creates 2^d exact, geometrically protected translated-and-rescaled copies of h. If h is computed by a ReLU network, the affine output of S merges into its first affine map, so this precomposition adds exactly one hidden layer.

Proof. Invoke Lemma A.1 in dimension d to obtain the small displacement map that will perturb the cube centers into quantitative affine general position. More precisely, Item 1 supplies a map Δ and defines the centers by

$$\boldsymbol{q_s} := \boldsymbol{s} + \Delta(\boldsymbol{s}), \qquad \boldsymbol{s} \in \left\{-\frac{1}{2}, \frac{1}{2}\right\}^d.$$

We first choose the side length r so that no affine hyperplane can meet more than d of the centered cubes, and then construct S so that it maps every such cube, after rescaling, onto $[-1,1]^d$. Define the unperturbing map

$$U(\boldsymbol{x}) := \boldsymbol{x} - \Delta(\boldsymbol{x}).$$

The subtraction reverses the local displacement: near $\boldsymbol{q_s}$, the map Δ still equals $\Delta(\boldsymbol{s})$, so subtracting it recovers the corresponding unperturbed point. Item 2 gives

$$\mathrm{Lip}(U) \le 1 + \mathrm{Lip}(\Delta) \le \frac{17}{16}.$$

For all distinct $\boldsymbol{s}_0, \dots, \boldsymbol{s}_d \in \left\{-\frac{1}{2}, \frac{1}{2}\right\}^d$, Item 4 gives

$$|\det Q(\boldsymbol{s}_0, \dots, \boldsymbol{s}_d)| \ge 2^{-C_1 d^3},$$

for a universal constant $C_1 > 0$, where $Q(\boldsymbol{s}_0, \dots, \boldsymbol{s}_d)$ is the $d \times d$ matrix whose i^{th} column is $\boldsymbol{q}_{\boldsymbol{s}_i} - \boldsymbol{q}_{\boldsymbol{s}_0}$.

There is a universal constant $C_2 > 0$ such that, for all $d \ge 2$,

$$2d\sqrt{d}\,(2\sqrt{d})^{d-1} \le 2^{C_2 d \log_2(d+1)}.$$

Choose

$$r := 2^{-C_0 d^3},$$

where $C_0 > C_1 + C_2$ is also large enough that $r < \frac{1}{8d}$. Since $d \log_2(d+1) \le d^3$ for $d \ge 2$, this choice gives

$$d(2r\sqrt{d})(2\sqrt{d})^{d-1} \le 2^{-(C_0 - C_2)d^3} < 2^{-C_1 d^3}. \tag{19}$$

The cubes

$$\boldsymbol{q_s} + [-r, r]^d, \qquad \boldsymbol{s} \in \left\{-\frac{1}{2}, \frac{1}{2}\right\}^d,$$

are pairwise disjoint and contained in $(-1,1)^d$. Indeed, distinct unperturbed vertices differ by one in some coordinate, whereas each center moves by at most $\frac{1}{64d}$ in that coordinate. Thus the centers $\boldsymbol{q_s}$ of any two distinct perturbed cubes are at ℓ_∞-distance greater than $\frac{3}{4}$, while $r < \frac{1}{8d}$. The same bounds put each of these cubes inside $(-1,1)^d$.

We now argue that no affine hyperplane intersects more than d of these cubes. Suppose by contradiction that one hyperplane meets cubes indexed by distinct $\boldsymbol{s}_0, \dots, \boldsymbol{s}_d$. Choose intersection points

$$\boldsymbol{v}_i := \boldsymbol{q}_{\boldsymbol{s}_i} + \boldsymbol{\xi}_i, \qquad \|\boldsymbol{\xi}_i\|_\infty \le r, \qquad \|\boldsymbol{\xi}_i\|_2 \le r\sqrt{d}, \qquad i = 0, \dots, d,$$

and let $V \in \mathbb{R}^{d\times d}$ be the matrix whose i^{th} column is $\boldsymbol{v}_i - \boldsymbol{v}_0$. Since all the $\boldsymbol{v}_i$ lie on one affine hyperplane,

$$\det V = 0.$$

Let $Q \coloneqq Q(\boldsymbol{s}_0, \dots, \boldsymbol{s}_d)$. The displacement bound and $\|\boldsymbol{\xi}_i\|_\infty \le r$ give

$$\|\boldsymbol{q}_{\boldsymbol{s}}\|_\infty \le \frac{1}{2} + \frac{1}{64d} < 1, \qquad \|\boldsymbol{v}_i\|_\infty \le \frac{1}{2} + \frac{1}{64d} + r < 1.$$

Every column of Q or V is therefore the difference of two vectors whose coordinates have absolute value less than one, and hence has Euclidean norm at most $2\sqrt{d}$.

For $i = 0, \dots, d$, let $Q^{(i)}$ be the matrix whose first i columns are those of V and whose remaining columns are those of Q. Thus $Q^{(0)} = Q$ and $Q^{(d)} = V$, so the triangle inequality gives

$$|\det Q - \det V| \le \sum_{i=1}^{d} \left|\det Q^{(i-1)} - \det Q^{(i)}\right|.$$

For each i, the only changed column is

$$(\boldsymbol{v}_i - \boldsymbol{v}_0) - (\boldsymbol{q}_{\boldsymbol{s}_i} - \boldsymbol{q}_{\boldsymbol{s}_0}) = \boldsymbol{\xi}_i - \boldsymbol{\xi}_0, \qquad \|\boldsymbol{\xi}_i - \boldsymbol{\xi}_0\|_2 \le 2r\sqrt{d}.$$

By multilinearity,

$$\det Q^{(i)} - \det Q^{(i-1)} = \det\left[\boldsymbol{v}_1 - \boldsymbol{v}_0, \dots, \boldsymbol{v}_{i-1} - \boldsymbol{v}_0, \boldsymbol{\xi}_i - \boldsymbol{\xi}_0, \boldsymbol{q}_{\boldsymbol{s}_{i+1}} - \boldsymbol{q}_{\boldsymbol{s}_0}, \dots, \boldsymbol{q}_{\boldsymbol{s}_d} - \boldsymbol{q}_{\boldsymbol{s}_0}\right].$$

Hadamard's inequality and the $2\sqrt{d}$ bound on every other column now give

$$\left|\det Q^{(i-1)} - \det Q^{(i)}\right| \le (2r\sqrt{d})(2\sqrt{d})^{d-1}.$$

Combining these estimates with Item 4, $\det V = 0$, and Equation (19), we obtain the contradiction

$$2^{-C_1 d^3} \le |\det Q| = |\det Q - \det V| \le d(2r\sqrt{d})(2\sqrt{d})^{d-1} < 2^{-C_1 d^3}.$$

We now define the copier. Let

$$\tau(t) \coloneqq -\frac{1}{2} + 2\left[t + \frac{1}{4}\right]_+ - 2\left[t - \frac{1}{4}\right]_+,$$

and use the same symbol for its coordinatewise application to vectors.

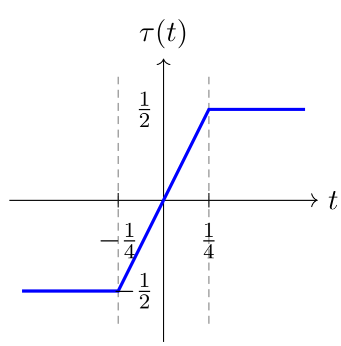

Figure 6: The scalar clipping function τ. It equals $-\frac{1}{2}$ on $\left(-\infty, -\frac{1}{4}\right]$, equals $\frac{1}{2}$ on $\left[\frac{1}{4}, \infty\right)$, and interpolates linearly between the two plateaus. The two axes use the same scale.

Set

$$S(\boldsymbol{x}) \coloneqq \frac{U(\boldsymbol{x}) - \tau(\boldsymbol{x})}{r}.$$

For all $\boldsymbol{s} \in \left\{-\frac{1}{2}, \frac{1}{2}\right\}^d$ and $\boldsymbol{u} \in [-1,1]^d$, Item 3 of Lemma A.1 gives

$$\Delta(\boldsymbol{q_s} + r\boldsymbol{u}) = \Delta(\boldsymbol{s}),$$

because $\|r\boldsymbol{u}\|_\infty \le r < \frac{1}{8d}$. Moreover,

$$|(\boldsymbol{q_s} + r\boldsymbol{u})_i| \ge \frac{1}{2} - \frac{1}{64d} - r > \frac{1}{4},$$

and its sign is the sign of s_i. Hence

$$\tau(\boldsymbol{q_s} + r\boldsymbol{u}) = \boldsymbol{s}.$$

Using Item 1 once more, we obtain

$$\begin{aligned} U(\boldsymbol{q_s} + r\boldsymbol{u}) &= \boldsymbol{q_s} + r\boldsymbol{u} - \Delta(\boldsymbol{q_s} + r\boldsymbol{u}) \\ &= \boldsymbol{s} + \Delta(\boldsymbol{s}) + r\boldsymbol{u} - \Delta(\boldsymbol{s}) \\ &= \boldsymbol{s} + r\boldsymbol{u}. \end{aligned}$$

Consequently,

$$S(\boldsymbol{q_s} + r\boldsymbol{u}) = \frac{U(\boldsymbol{q_s} + r\boldsymbol{u}) - \tau(\boldsymbol{q_s} + r\boldsymbol{u})}{r} = \frac{\boldsymbol{s} + r\boldsymbol{u} - \boldsymbol{s}}{r} = \boldsymbol{u}.$$

It follows that, for all functions h,

$$(h \circ S)(\boldsymbol{q_s} + r\boldsymbol{u}) = h(\boldsymbol{u}).$$

It remains to verify that the claimed architecture can be implemented using a single hidden layer with the claimed width. In one common hidden layer, concatenate three feature blocks: the $\mathcal{O}(d^4)$ ReLU features used by the depth-2 realization of Δ; the $2d$ neurons

$$\left[x_i + \frac{1}{4}\right]_+, \quad \left[x_i - \frac{1}{4}\right]_+, \qquad i \in [d],$$

that compute the d coordinates of $\tau(\boldsymbol{x})$; and the $2d$ identity carriers

$$[x_i]_+, \quad [-x_i]_+, \qquad i \in [d],$$

from which the affine output recovers $x_i = [x_i]_+ - [-x_i]_+$. Every neuron in these three blocks applies a ReLU to an affine function of the same input $\boldsymbol{x}$, so all blocks are evaluated in parallel by this single hidden layer. Its affine output computes, for all $i \in [d]$,

$$S_i(\boldsymbol{x}) = \frac{x_i - \Delta_i(\boldsymbol{x}) - \tau(x_i)}{r}.$$

The total width is $\mathcal{O}(d^4) + 4d = \mathcal{O}(d^4)$. Finally, the coordinatewise map τ is 2-Lipschitz, and therefore

$$\mathrm{Lip}(S) \le \frac{\mathrm{Lip}(U) + \mathrm{Lip}(\tau)}{r} < \frac{4}{r}.$$

$\square$

Proof of Theorem 3.1. Fix r, $\{\boldsymbol{q_s}\}$, and S supplied by Proposition C.2, and let h_0 and $\mathcal{D}_0^d$ be the affine base pair from Lemma C.1. Let $\boldsymbol{X}_0 \sim \mathcal{D}_0^d$. For $j = 1, \ldots, \ell - 2$, set

$$h_j \coloneqq h_{j-1} \circ S.$$

Define the matching random vector $\boldsymbol{X}_j$ recursively: at each step, independently of $\boldsymbol{X}_{j-1}$, choose $\boldsymbol{s}$ uniformly from $\left\{-\frac{1}{2}, \frac{1}{2}\right\}^d$ and set $\boldsymbol{X}_j \coloneqq \boldsymbol{q_s} + r\boldsymbol{X}_{j-1}$. Equivalently, for all bounded measurable functions F,

$$\mathbb{E}[F(\boldsymbol{X}_j)] = 2^{-d} \sum_{\boldsymbol{s} \in \left\{-\frac{1}{2}, \frac{1}{2}\right\}^d} \mathbb{E}\left[F(\boldsymbol{q_s} + r\boldsymbol{X}_{j-1})\right]. \tag{20}$$

Here j counts copier layers: h_j is the target after j copying steps, while $\boldsymbol{X}_j$ makes the same j choices of copies. Item 3 of Proposition C.2 gives, for all $\boldsymbol{u} \in [-1, 1]^d$,

$$h_j(\boldsymbol{q_s} + r\boldsymbol{u}) = h_{j-1}\big(S(\boldsymbol{q_s} + r\boldsymbol{u})\big) = h_{j-1}(\boldsymbol{u}). \tag{21}$$

Thus each recursion step replaces the preceding target-distribution pair by 2^d exact, equally weighted copies of it that are slightly perturbed.

We first prove the separation for these undilated pairs whose support is inside the unit hypercube. We claim that, for all $j = 0, \ldots, \ell - 2$, all depth-$(j+1)$ networks N of width at most w, where $wd \leq 2^d$, satisfy

$$\mathbb{E}\left[\big(N(\boldsymbol{X}_j) - h_j(\boldsymbol{X}_j)\big)^2\right] \geq \frac{1}{12}\left(1 - \frac{wd}{2^d}\right)^j. \tag{22}$$

At $j = 0$, the competitor has depth 1 and is therefore affine, so the claim is exactly Lemma C.1.

For the induction step, suppose the claim holds at $j - 1$ and let N have depth $j + 1$. Its first layer has at most w genuine activation hyperplanes. By Item 2 of Proposition C.2, each such hyperplane meets at most d of the 2^d protected cubes. Consequently, at least $2^d - wd$ of the outer cubes $\boldsymbol{q_s} + r(-1, 1)^d$, each carrying one copy of h_{j-1}, are missed by every first-layer hyperplane.

On a missed cube, every non-constant first-layer preactivation has a fixed sign. A constant preactivation also has an affine ReLU output: it is either a constant or identically zero. Hence every first-layer neuron is affine throughout that cube, and the entire first hidden layer can be absorbed into the next affine map. For the corresponding $\boldsymbol{s}$, the function

$$\boldsymbol{u} \longmapsto N(\boldsymbol{q_s} + r\boldsymbol{u})$$

therefore agrees on $(-1, 1)^d$ with a depth-j, width-at-most-w network. The induction hypothesis and Equation (21) imply that its conditional squared error is at least

$$\frac{1}{12}\left(1 - \frac{wd}{2^d}\right)^{j-1}.$$

The distribution in Equation (20) gives each outer copy mass 2^{-d}. Keeping the errors on the missed copies and discarding the non-negative errors on all other copies yields

$$\mathbb{E}\left[\big(N(\boldsymbol{X}_j) - h_j(\boldsymbol{X}_j)\big)^2\right] \geq \frac{2^d - wd}{2^d} \cdot \frac{1}{12}\left(1 - \frac{wd}{2^d}\right)^{j-1} = \frac{1}{12}\left(1 - \frac{wd}{2^d}\right)^j.$$

This proves Equation (22).

We now dilate the undilated pair to obtain the pair stated in the theorem. Since h_0 has depth 2, width 3, and Lipschitz constant one, $h_{\ell-2}$ has depth ℓ, width $\mathcal{O}(d^4)$, size $\mathcal{O}(\ell d^4)$, and

$$\mathrm{Lip}(h_{\ell-2}) \leq \left(\frac{4}{r}\right)^{\ell-2}.$$

Let

$$\alpha := 4^{\ell-2} r^{-(\ell-2)}, \qquad f_{\ell,d}(\boldsymbol{x}) := h_{\ell-2}\left(\frac{\boldsymbol{x}}{\alpha}\right).$$

We now specify the distribution $\mathcal{D}_\ell^d$ appearing in Theorem 3.1: sample $\boldsymbol{X}_{\ell-2}$ as above and output $\alpha\boldsymbol{X}_{\ell-2}$. The input dilation changes neither the depth, width, nor size of the realizing network. Since h_0 takes values in $[0,1]$, so does every h_j and hence $f_{\ell,d}$. Moreover,

$$\mathrm{Lip}(f_{\ell,d}) \leq \frac{1}{\alpha}\left(\frac{4}{r}\right)^{\ell-2} = 1.$$

The same dilation leaves squared approximation error unchanged; this is also the input-dilation principle in [23, Theorem 9]. Explicitly, for all networks N,

$$\mathop{\mathbb{E}}_{\boldsymbol{x}\sim\mathcal{D}_\ell^d}\left[\big(N(\boldsymbol{x}) - f_{\ell,d}(\boldsymbol{x})\big)^2\right] = \mathbb{E}\left[\big(N(\alpha\boldsymbol{X}_{\ell-2}) - h_{\ell-2}(\boldsymbol{X}_{\ell-2})\big)^2\right]. \tag{23}$$

Let N now be a depth-$(\ell-1)$ network satisfying the width bound in the theorem. Bernoulli's inequality gives

$$\left(1 - \frac{wd}{2^d}\right)^{\ell-2} \geq 1 - (\ell-2)\frac{wd}{2^d} \geq \frac{1}{2}.$$

The network $\boldsymbol{u} \mapsto N(\alpha\boldsymbol{u})$ has the same depth and width as N. Equations (22) and (23) therefore give

$$\mathop{\mathbb{E}}_{\boldsymbol{x}\sim\mathcal{D}_\ell^d}\left[\big(N(\boldsymbol{x}) - f_{\ell,d}(\boldsymbol{x})\big)^2\right] \geq \frac{1}{24}.$$

This proves the two depth-separation claims.

We finish by verifying the additional distributional properties stated after the theorem. None of them is used in the lower-bound induction above. The number, size, and location of the supporting cubes make the geometry and exponential spread of the distribution explicit. The conditional-density bound is recorded specifically for comparison with the Vardi–Shamir barrier [27].

Induction in Equation (20) shows that the law of $\boldsymbol{X}_j$ is uniform on 2^{dj} pairwise disjoint open cubes of side length $2r^j$, all contained in $(-1,1)^d$. Indeed, at the next step the map $\boldsymbol{x} \mapsto \boldsymbol{q}_{\boldsymbol{s}} + r\boldsymbol{x}$ shrinks every old cube by r; branches with different $\boldsymbol{s}$ lie in the disjoint protected cubes $\boldsymbol{q}_{\boldsymbol{s}} + r(-1,1)^d$, while cubes within one branch remain disjoint. The equal branch weights and equal Jacobians preserve uniformity.

To locate these cubes relative to the origin, unroll the recursion. For $j \geq 1$, all $\boldsymbol{x}$ in the support of the law of $\boldsymbol{X}_j$ have the form

$$\boldsymbol{x} = \boldsymbol{q}_{\boldsymbol{s}_1} + r\boldsymbol{q}_{\boldsymbol{s}_2} + \cdots + r^{j-1}\boldsymbol{q}_{\boldsymbol{s}_j} + r^j\boldsymbol{u}$$

for some $\boldsymbol{s}_1, \ldots, \boldsymbol{s}_j \in \left\{-\frac{1}{2}, \frac{1}{2}\right\}^d$ and $\boldsymbol{u} \in [-1,1]^d$. Every coordinate of $\boldsymbol{q}_{\boldsymbol{s}_1}$ has magnitude at least $\frac{1}{2} - \frac{1}{64d}$, whereas $\left\|\boldsymbol{q}_{\boldsymbol{s}_k}\right\|_\infty < \frac{5}{8}$. Since $r < \frac{1}{8d} \leq \frac{1}{16}$ and $r^j \leq r$,

$$\|\boldsymbol{x}\|_\infty \geq \frac{1}{2} - \frac{1}{64d} - \frac{5r}{8(1-r)} - r^j > \frac{1}{2} - \frac{1}{128} - \frac{1}{24} - \frac{1}{16} > \frac{1}{8}.$$

The final dilation turns each cube supporting the law of $\boldsymbol{X}_{\ell-2}$ into a cube of side length

$$2\alpha r^{\ell-2} = 2 \cdot 4^{\ell-2}.$$

Because the undilated support lies in $[-1,1]^d$ and outside $[-\frac{1}{8}, \frac{1}{8}]^d$, the support of $\mathcal{D}_\ell^d$ satisfies

$$\inf_{\boldsymbol{x} \in \operatorname{supp}(\mathcal{D}_\ell^d)} \|\boldsymbol{x}\|_2 \geq \frac{\alpha}{8}, \qquad \sup_{\boldsymbol{x} \in \operatorname{supp}(\mathcal{D}_\ell^d)} \|\boldsymbol{x}\|_2 \leq \sqrt{d}\,\alpha.$$

After adjusting universal constants, these bounds are respectively $2^{c(\ell-2)d^3}$ and $2^{C(\ell-2)d^3}$. For almost every fixed value of the other $d-1$ coordinates, a coordinate fiber meets some $k \geq 1$ supporting cubes in disjoint intervals, each of length $2 \cdot 4^{\ell-2}$. Its conditional law is uniform on that union, so its density is

$$\frac{1}{2 \cdot 4^{\ell-2} k} \leq \frac{1}{2}.$$

This completes the proof. $\square$